OpenFX

The FX Iceberg



Report
Q3 2026

openfx.com

INTRODUCTION

Why Liquidity?

Liquidity is often treated as a simple question: is there enough money available to transact?

When discussing the cross-border payments industry, the answer to that question is rarely sufficient. Whether fiat is technically available says little about whether it's transferable.

A currency can be liquid and still be difficult to trade at the sizes you need.

A quote can seem competitive but disappear before you can execute the trade.

A transaction can be traveling through a nominally fast corridor but still take far longer than it should because of exogenous conditions.

This report uses liquidity as the lens for understanding the FX market more generally, and the payments market in particular. In it, we expand on an idea that is fundamental to how we do business: if you want to know whether money can move efficiently across a corridor, you have to look beyond nominal asset availability and ask about the surrounding machinery that supports the transaction.

Performance in this framing is context-dependent. The same market can behave differently as transaction size rises or exogenous stress enters the picture.

The ascendance of stablecoins makes this kind of analysis more critical. These rails can make it easy to forget that a wallet-to-wallet transfer is not a cross-border payment. While they can dramatically speed up portions of the process, they cannot fully remove market complexities, only relocate them.

The ability to move money reliably across the world, whether through traditional FX or stablecoin-powered rails, always comes back to high-quality liquidity, and high-quality liquidity is a question of operations and infrastructure.

We make this argument in three parts. Act I looks at how liquidity shapes the market and why FX operates so differently across the world. Act II asks how high-quality liquidity should actually be measured. Act III turns to the builder's side and examines what has to exist underneath the transaction to produce those outcomes reliably.

We do this using a framework we developed called the FX Iceberg, designed to show the interdependence of these functions and offer a mental model for determining why a certain transaction under certain conditions might fail.

The goal is not to replace one simplified number with another; it is to give the reader a better way to ask what makes a corridor work, how to tell when it does, and what has to be built when it does not.

Can the system move the amount required, at the expected price, and deliver usable money reliably?

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The Current State of FX



01

I.01

2026 FX Fact Book

How big is the cross-border FX market?

That depends on what you're asking.

If the question is how much total value moved across borders, FXC Intelligence estimates that in 2025 there were about $208 trillion in cross-border payment flows. Roughly $164 trillion of that was wholesale, and the rest retail.[1]

Move inward to the currency market itself, and in April 2025 the Bank for International Settlements (BIS) measured average daily OTC FX turnover to be $9.6 trillion.[2] Follow those trades further into settlement and the figure changes again: more than $14 trillion in gross FX obligations per day.[3]

Let's add stablecoins into the mix, which have become a notable source of flows. BIS put stablecoin market capitalization at roughly $315 billion in early April 2026. In 2025, the market generated about $35 trillion worth of transaction activity, with roughly $390 billion classified as payment-related flows.[4]

Each number seeks to answer a different question about the same broad system. Change the question and the number changes alongside it.

So in any discussion about the FX market, you must start with the questions you plan to pursue. This report seeks to answer one in particular: “What prevents the global payments market from operating more smoothly?” We seek to answer it through the lens of liquidity, and through a framework we are building around measuring the quality of that liquidity across the lifecycle of a cross-border transaction.

Before we begin, let's take a survey of the market's other numbers.

EXHIBIT A1

2026 FX Fact Book

LAYER	MEASURE	LATEST FIGURE	PERIOD	WHAT IT MEASURES	BOUNDARY / CONTEXT
Payments	Cross-border payment flow	$208T / year[1]	2025	Estimated value of global cross-border payments	FXC model: $164T wholesale + $44T retail. Separate modeled payment-provider revenue pool: $625B.
FX trading	OTC FX turnover	$9.6T / day[2]	Apr 2025	Average daily OTC FX trading turnover	$3.0T spot; $1.8T outright forwards; $4.0T FX swaps.
FX trading	Dollar share	89.2%[2]	Apr 2025	Share of FX trades with USD on one side	Every one of the ten most-traded currency pairs involved USD.
FX trading	Trading-center concentration	75%[2]	Apr 2025	Share of global turnover intermediated by sales desks in the UK, US, Singapore and Hong Kong SAR	Location of reporting sales desks; this is a geography-of-intermediation measure.
FX trading	Participant mix	50% / 46%[2]	Apr 2025	Other financial institutions / interdealer shares of global turnover	Aggregate counterparty categories.

EXHIBIT A1

LAYER	MEASURE	LATEST FIGURE	PERIOD	WHAT IT MEASURES	BOUNDARY / CONTEXT
Settlement	Gross FX settlement obligations	>$14T / day[3]	Apr 2025	Average daily gross financial obligations generated by FX settlement	Settlement obligations are measured separately from trading turnover.
Stablecoins	Market capitalization	~$315B[4]	Early Apr 2026	Outstanding stablecoin value	Stock measure.
Stablecoins	Transaction activity	~$35T / year[4]	2025	Total stablecoin transaction volume	Broad activity measure; much of the activity sits inside the crypto ecosystem.
Stablecoins	Payment-related activity	~$390B / year[4]	2025	Estimated payment-related stablecoin flows	Narrow subset of total stablecoin transaction activity.
Stablecoins	USD denomination	~98%[4]	2026 context	Stablecoin value denominated in US dollars	Denomination measure.

Let's begin with $9.6 trillion per day. An extraordinary amount of money is changing hands, being converted from one currency to another, being settled through hundreds of different regulatory and banking systems. The quality of those movements varies enormously around the world, and liquidity explains much of that variation.

In fact, USD appeared on one side of 89.2% of trades in April 2025, and every one of the ten most- traded currency pairs involved the dollar.[2]

Why does so much of the market run through the dollar?

Many currency pairs are not liquid enough to support efficient direct conversion and rely on a “vehicle” to bridge them, usually USD, the most liquid currency on Earth.[5]

This lack of direct liquidity sits at the center of almost every one of the market pathologies in cross- border FX. It can slow transactions, raise fees, and introduce additional risk for both the banks involved and their counterparties.

This is the cost of low-quality liquidity.

I.02

Inside the $9.6 Trillion Market

Institutional FX is decentralized. There is no single exchange where all global demand arrives and clears. A client can arrange a quote with a dealer on the phone, stream prices from one dealer through Bloomberg, trade through an electronic network or order book,[1] or through a Telegram chat. The choice depends on the currency being traded, the size of the transaction, and yes, liquidity.

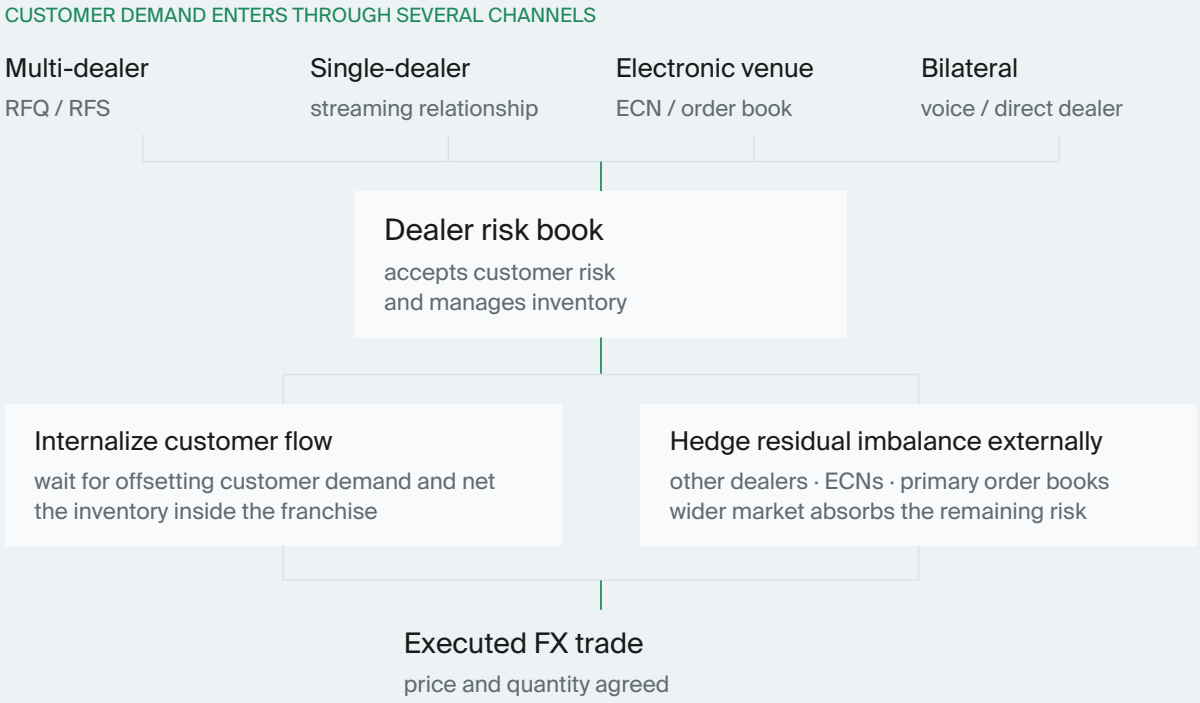
Whatever the route, a dealer taking the other side of the trade inherits the underlying currency risk. The first question they must ask is how much to internalize, and how much to hedge externally. This too often comes down to liquidity.

For example, a dealer can accept a customer trade and wait for offsetting demand from another customer rather than immediately hedging the position on the wider market. A euro purchase from one client can be offset by selling euros to another, allowing the dealer to recycle risk inside its own book. The largest dealers must make these decisions at enormous scale. BIS estimates customer- spot internalization at roughly 84% in the United Kingdom and United States, and about 78% in Hong Kong SAR and Singapore in 2025.[2]

Internalization turns the dealer's customer base into a source of liquidity. A broad mix of buying and selling gives the dealer more chances to offset positions internally. When enough offsetting flow does not appear, the remaining inventory has to be managed elsewhere, through other dealers or the market itself.[3]

EXHIBIT A2

How Institutional FX Absorbs Demand



Execution is only half the machinery

Whatever the decision, once the necessary trades are executed, the two currency obligations still have to be discharged through settlement. The speed and risk of settlement are determined by the machinery that sits between the two institutions.

For trades settled through CLSSettlement, for example, settlement is handled using payment versus payment (PvP), which can remove principal settlement risk by ensuring both sides of the transaction settle simultaneously. Where PvP settlement is unavailable, institutions must reduce exposure through netting, other intragroup structures, or carefully controlling the timing of transactions. In some cases, the risk cannot be completely eliminated. It's estimated that in 2025, $1.4 trillion a day still settled bilaterally, fully exposed to settlement risk. This represented about 10% of the market.[4]

All of this, and the complications that sit between, can be understood as functions of liquidity: how much currency is available to complete transactions or hedge the resulting risk.

In a low-liquidity regime, you might not be able to complete a transaction at any price, but even in a technically high-liquidity regime, there still might be limits to the scale of the transaction that can be completed.

I.03

Liquidity Depends on Transaction Size

How liquid is a currency?

Before answering, you need one more number: how much are you trying to trade?

Transaction size changes the amount of market capacity an order needs. A $500 currency conversion at an airport kiosk might consume only a sliver of available liquidity; filling a $25 million treasury order can require substantially more, even in the same currency pair at the same moment.

That makes transaction size part of the definition of liquidity. Clean execution at hundreds or thousands of dollars cannot establish the execution quality of a $10 million or $25 million treasury trade. This distinction matters because sub-institutional evidence is often laundered to make institutional claims. A liquidity provider might be able to easily move a million dollars, but fall apart as clip sizes rise.

What happens when an order gets larger?

On an order book, the best available price covers a finite amount. Once an order consumes that quantity, the remainder moves to the next available price and continues through the book until the full order is filled. Sweep-to-fill measures the price paid across that path. When liquidity is shallow, larger orders have farther to travel.

A historical study of EBS and Reuters order books gives us a look into the mechanism. It calculated sweep-to-fill half-spreads for $1 million, $10 million and $25 million aggressive orders across selected currencies from January 2014 through March 2018.[1]

TRANSACTION SIZE $1M $5M $50M

To understand liquidity you must first understand transaction size.

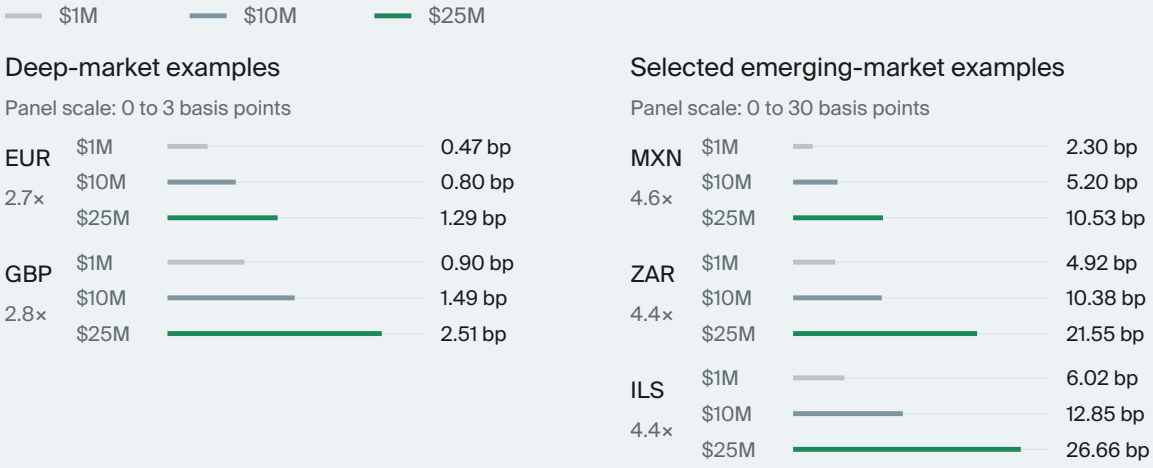
A transaction that can be executed effectively at $1M or $5M dollars might fall apart at $50M.

50M

EXHIBIT A3

Size Changes the Price You Can Actually Execute

Historical EBS / Reuters sample, 2 Jan 2014 to 22 Mar 2018. Sweep-to-fill half-spread in basis points.



Depth can become the constraint itself. In several emerging-market observations in the study, the displayed EBS / Reuters order book did not contain $25 million of depth. The authors extrapolated the $25 million cost estimate in those cases.[1]

As you can see, price is as much a function of clip size as it is a function of currency. These two factors reinforce one another in the market. An illiquid currency like ZAR is inherently more expensive than a G7 currency like EUR, and that expense will increase faster as trade volume increases.

Both are a function of illiquidity itself. As the volume becomes harder to fill and the book becomes thinner, the price premium for every additional dollar increases.

This is why scale discipline matters. Understanding how a hundred-dollar remittance payment or a $20,000 B2B payment operates doesn't tell you how a $25 million treasury order will be filled.

Institutional liquidity has to be tested with institutional demand.

I.04

The Two Liquidity Worlds

Early on, we noted that the FX market is actually multiple markets, depending on the lens you view it through and the questions you're asking.

The lens we have chosen for this report is liquidity, because our argument is that liquidity sits beneath almost everything else.

Liquidity explains settlement speed, pricing, who gets to move money, and how those flows get moved.

Up to now we've offered multiple points of evidence for this claim. Liquidity determined how much customer demand a dealer could absorb before it needed the wider market, and we saw the same pressure in settlement, hedging, and transaction size.

Seen through this lens, the $9.6 trillion-a-day global market begins to separate. Some currencies operate in a regime where settlement is fast, clip size is less binding, and deep hedge markets make the residual risk easier to move. These markets can remain stable during periods of relative crisis, and have the infrastructure to support reliability across a range of market conditions.[3]

Other currency corridors are not as robust. Flow can be more directional, leaving dealers with inventory that is harder to offset. The hedge itself can be shallow or split across venues. FX and capital controls can fracture activity between onshore and offshore markets.[1]

These differences accumulate. We call the environments they create the High Liquidity World and the Low Liquidity World.

The High Liquidity World describes markets where substantial institutional demand can find enough natural flow, balance sheet, hedging capacity, and settlement access to remain comparatively predictable as size rises.

The Low Liquidity World begins where those buffers begin to fail. The market can still function, sometimes at considerable scale, but it is far less robust. The consequences appear in price, available depth, execution method, hedging choices, funding, and settlement.

THESE DIFFERENCES ACCUMULATE.

High Liquidity WORLD

Low Liquidity WORLD

EXHIBIT A4

Two Liquidity Worlds

ENVIRONMENT 1	ENVIRONMENT 2
High Liquidity World	**Low Liquidity World**
More capacity is available before price and execution conditions begin to deteriorate.	Capacity binds sooner, so larger demand exposes friction earlier.
CUSTOMER FLOW Dense, varied flow creates more opportunities for natural offsets.	CUSTOMER FLOW More directional or concentrated flow leaves fewer natural offsets.
DEALER CAPACITY Dealers can absorb more imbalance internally or warehouse it for longer.	DEALER CAPACITY Inventory becomes harder to carry and external hedging pressure arrives sooner.
HEDGING Deeper spot and derivative markets give residual risk more places to go.	HEDGING Shallower or fragmented hedge markets make residual risk more expensive to move.
MARKET ACCESS Broad deliverability and connected venues keep more liquidity economically usable.	MARKET ACCESS Deliverability restrictions can divide liquidity across onshore and offshore markets.[1]
SETTLEMENT Mature wholesale infrastructure provides broader access to risk-reducing settlement methods.	SETTLEMENT Currency eligibility, operating access, and local infrastructure can narrow available settlement paths.[2]
AT SIZE Execution conditions tend to degrade later as demand rises.	AT SIZE Price, depth, and execution conditions can deteriorate sooner.

One visible infrastructure difference: payment-versus-payment usage varies sharply with currency-pair eligibility. This is one settlement dimension inside the broader liquidity picture.

40% CLS-eligible pairs

12% non-eligible pairs

Market conditions, not country classifications.

The chart outlines the various conditions that distinguish these regimes. None of this is set in stone. No currency is condemned to exist in the “Low Liquidity World,” and many of the countries with thin liquidity have worked hard to improve the underlying conditions that drive these pathologies.

The same currency can move closer to one world or the other depending on the product, venue, corridor, requested size, and moment in the market. Country membership is too coarse to capture those changes.

With this in hand, we can start to look at these worlds more closely.

I.05

Exploring The Low Liquidity World

What actually makes a market illiquid?

Up to this point, we have described the Low Liquidity World primarily in terms of effects: wider execution costs, thinner depth, greater sensitivity to size, harder hedging, and less predictable settlement. Those effects tell us what thin liquidity feels like once a transaction reaches the market. They tell us much less about where the friction came from.

Convertibility rules, dealer capacity, hedge markets, market access, settlement infrastructure, local funding, and payment systems all determine how much currency is actually available to complete or hedge a transaction, and where that capacity runs out. While the effects are often predictable, there is no universal shape.

Once you look underneath the effects, the differences become obvious.

INTERLOCKING SYSTEMS

The useful question is where the systems start to break down.

Every low-liquidity regime is built on top of interlocking systems that add friction.

EXHIBIT A5

Five Systems Beneath Thin Liquidity

LATIN AMERICA **Colombia** INSTITUTIONAL FX MARKET STRUCTURE	WHAT ADDS FRICTION Spot trading is liquid and competitive, while hedging is concentrated in NDFs and nonresident COP settlement remains restricted.	→	WHAT IT DOES TO LIQUIDITY Spot can trade like the High Liquidity World while hedging and settlement remain harder.[1]
SUB-SAHARAN AFRICA **Ethiopia & Nigeria** FX REFORM AND MARKET FUNCTIONING	WHAT ADDS FRICTION Ethiopia combined residual restrictions, capital controls, and a shallow financial market. Nigeria has been rebuilding interbank price discovery and market infrastructure.	→	WHAT IT DOES TO LIQUIDITY Restrictions pushed demand out of the official market and weakened price discovery. Reforms improved access to FX and interbank liquidity, though some of Ethiopia's underlying frictions later reappeared.[2][3]
MIDDLE EAST & NORTH AFRICA **Egypt** FX ACCESS, BACKLOG AND INTERBANK TURNOVER	WHAT ADDS FRICTION A segmented exchange-rate system and constrained FX availability had pushed demand into a parallel market and created bank backlogs.	→	WHAT IT DOES TO LIQUIDITY FX became easier to source through banks: the parallel-rate gap closed, backlogs cleared, and interbank trading increased.[4]
SOUTH ASIA **India** DELIVERABILITY AND DERIVATIVE MARKET STRUCTURE	WHAT ADDS FRICTION Restrictions on offshore deliverability split INR activity between onshore and offshore markets.	→	WHAT IT DOES TO LIQUIDITY Hedging becomes harder because liquidity is split between onshore and offshore markets, and access depends on where the participant can trade.[5]
EAST ASIA **South Korea** MARKET ACCESS, TRADING HOURS AND TURNOVER	WHAT ADDS FRICTION Foreign institutions historically had limited direct access to the onshore market, trading hours were short, and KRW activity leaned heavily on offshore NDFs.	→	WHAT IT DOES TO LIQUIDITY Broader access and longer trading hours gave foreign institutions more ways to reach the onshore market. Daily FX turnover rose 13% year over year in the first four months after the change and was 40% above the average of the previous five years.[6]

Turnover measures activity, not spread, depth, or execution quality.

Colombia shows how uneven those conditions can be inside a single market. Its spot market can operate competitively while hedging and settlement retain characteristics associated with the Low Liquidity World. Parts of a low-liquidity market can look almost identical to the High Liquidity World.[1]

The reformers tell us something else: liquidity regimes can move when the systems underneath them change. Policymakers in Ethiopia and Egypt unified exchange rates, Nigerian authorities rebuilt parts of the interbank FX market, and Korean authorities widened foreign access and trading hours. Each intervention targeted a different source of friction. Change the systems, and liquidity conditions can change with them.[2346]

These examples illustrate that the Low Liquidity World is a family of recurring constraints whose exact mix can change from market to market.

That also means there is no universal cure. The intervention that matters depends on where the friction originates. Opening access may matter most in one market; somewhere else, the binding problem may sit in the hedge market or exchange-rate system.

That variability is why our two worlds describe changing market conditions. Country identity is too coarse to capture them. A country can contain several liquidity environments at once, and the same currency can behave differently across products, corridors, sizes, and moments in the market.

Once the Low Liquidity World is understood this way, the next question is scale: how much of the world operates under some version of these conditions, and what do they cost?

I.06

The Low Liquidity World Is Most of the World

So how much of the world actually belongs to it?

We cannot give you an exact country count. Liquidity depth inside a country is multivariate, as we have previously shown, so trying to draw a hard border around the Low Liquidity World would be an exercise in futility.

What we can say with confidence is that most countries on Earth are, to a greater or lesser degree, part of the club. The deep-liquidity core is unusually small. One of the clearest proofs is hiding in a number we have already raised.

USD appeared on one side of 89.2% of global FX trades in April 2025.[1]

The dollar acts as the switchboard because the market needs a switchboard. A vehicle currency solves a problem created by uneven direct liquidity: the need to find buyers and sellers willing to execute a transaction at a reasonable price. While they might not be willing to trade KRW or COP directly, for instance, everyone is willing to trade USD.[2]

In this example, the COP-to-KRW trade can be broken into COP-to-USD and USD-to-KRW, two legs with higher relative liquidity. Thinner markets can reach one another through a deeper common pool.

Friction is revenue

There is another difference between the two worlds that might go some way toward explaining why this bifurcation persists. The deepest regional markets carry extraordinary amounts of payment flow while providers earn comparatively little on each dollar moved. Deep liquidity and mature infrastructure invite stronger competition, which compresses the commercial take rate toward zero.

The economics change as friction rises. FXC Intelligence's 2025 B2B model shows Latin America, Africa, and the Middle East generating a relatively massive share of revenues compared with volume. FXC attributes these differences to competition, volatility, operational and compliance friction, and liquidity.[3]

EXHIBIT A6

The Switchboard and the Revenue Pool

The dollar switchboard

Institutional FX market structure · April 2025

89.2%

of global FX trades had USD on one side.[1]

96%

of CNY FX transactions were against USD.[1]

COP → USD → KRW

Where the B2B revenue sits

Modeled cross-border B2B payment flow and provider revenue · 2025

REGION	FLOW	REVENUE	TAKE RATE
Asia-Pacific	31%	38%	0.8%
Europe	40%	31%	0.5%
Middle East	5%	8%	1.2%
North America	15%	6%	0.3%
Latin America & Caribbean	6%	12%	1.5%
Africa	3%	5%	1.1%

The revenue pattern gives the deepest regional markets a slightly strange economic character: they are incredibly popular and generate comparatively little provider revenue on each dollar moved.

Friction in this industry has economic value. The same conditions that make a corridor difficult for the customer create opportunities for institutions that can navigate those constraints.

Those economics inform the pace of reform. BIS has linked rising demand and greater concentration in correspondent banking to probable rent-seeking, and has separately noted that some private institutions have been cautious about changes that could disrupt their business models.[4][5]

Our contention from this and other evidence is straightforward: some cross-border frictions persist partly because they are profitable to intermediaries. This is obviously not the entire story, but the Low Liquidity World does not persist simply because no one has recognized the pattern.

We now have a market where most countries depend, to some degree, on liquidity concentrated elsewhere. The dollar provides the great common pool connecting those markets, and managing the disjunctions can itself generate revenue.

Which raises the next question: how did the global FX system come to be organized this way?

I.07

How The Liquidity Divide Came To Be

How did the global FX system come to be organized this way?

The modern FX system was assembled one problem at a time. Governments and financial institutions kept building around the constraint in front of them. Those solutions accumulated, and the architecture we use today still carries their shape.

Start with the dollar

The postwar system put USD at the center. Under Bretton Woods, participating currencies maintained fixed but adjustable rates to the dollar, while the dollar itself was fixed to gold. When convertibility became operational in 1958, countries settled international balances in dollars.[1]

That decision gave the expanding international system a common monetary hub. It also gave later market infrastructure a dollar-centered starting point.

Then trade outran geography

Global commerce created a practical problem: banks needed to send and receive money inside foreign banking systems where they had no direct presence. Correspondent banking supplied the reach. The network had existed for decades, but its scale and complexity expanded substantially with the late-twentieth-century wave of globalization.[2]

A bank could now reach another currency through a relationship with a bank that already lived inside that market. International access grew through those relationships, making them part of the architecture itself.

The day settlement risk became impossible to ignore

On 26 June 1974, Bankhaus Herstatt was liquidated after counterparties had already paid Deutsche marks. Its New York correspondent then stopped the outgoing dollar payments. One side of the trade had settled. The other never arrived.[3]

The industry had a vivid demonstration that executing an FX trade and safely completing it were separate problems. The response took decades. CLS launched in 2002, using payment versus payment so eligible transactions could settle both currency legs together and remove principal settlement risk from the covered trade.[3]

Markets route around the walls

Elsewhere, the problem was access to the currency itself. Capital controls, restricted convertibility, and thin domestic markets helped push hedging activity offshore. NDF markets let participants manage currency risk without requiring delivery of the underlying currency.[4]

The logic repeats. When institutions could not remove a constraint, they often built a way to transact around it.

Then the network thinned

The correspondent system that had extended global reach later began to contract. BIS documents a broad retreat in active correspondent relationships after decades of expansion.[2]

EXHIBIT A7

How The Liquidity Divide Came To Be

1944 → 1958	A dollar-centered system	Bretton Woods tied participating currencies to USD; once convertibility became operational, international balances were settled in dollars.[1]
LATE 20TH CENTURY	Correspondent reach expands	Globalization drove a larger, more complex network of bank relationships connecting otherwise separate national systems.[2]
1974	Herstatt exposes settlement risk	A bank failure leaves one currency leg paid and the other unpaid, making principal settlement risk impossible to ignore.[3]
HISTORICAL DEVELOPMENT	NDF markets grow offshore	Where currencies are difficult to deliver or domestic markets are thin, offshore NDFs create another way to hedge.[4]
2002	CLS adds PvP settlement	Eligible FX activity can settle both legs together, removing principal risk from the covered transaction.[3]
2010S → 2020S	Correspondent reach contracts	Active correspondent relationships retreat after decades of expansion, leaving a smaller network carrying global access.[2]

Arrive in 2025

$14T+	36%	$1.4T+	≈30%
average daily FX settlement obligations	settled using payment versus payment	gross bilateral settlement exposed to full principal risk	decline in active correspondents over the preceding decade in the latest statistics through 2022.[2]

By 2025, global FX had enormous reach alongside uneven access to its strongest safeguards.

Affected markets kept building to fix the shortfall. Central banks and regional institutions have created new payment rails and regional systems to improve the conditions they inherited.

Workarounds for a world not designed for them.

Which raises the next question: what have they built?

I.08

Building Around the Friction

Quite a lot.

The Low Liquidity World has not been waiting for the global financial system to become more accommodating. Central banks and regional institutions have been building around the constraints they inherited.

The work often begins at home, with faster domestic rails. While these systems cannot deepen an FX market themselves, they can remove delay once local currency is available.

Colombia's Bre-B is a great example. At launch in October 2025, it connected 227 institutions for 24/7 payments. In its first six months, the system processed more than 617 million transactions worth over COP 97 trillion.[6]

Other projects reach across borders. Africa built PAPSS. Buna serves the Arab region. Project Nexus is designed to connect instant-payment systems across six Asian markets. ASEAN governments have also expanded their own currency transaction frameworks.[2][4][5]

Different designs meeting the same problem: shorten the time and cost of settlement.

EXHIBIT A8

Workarounds for a World Not Designed for Them

World Bank FASTT tracks fast-payment-system availability globally.[1]

COLOMBIA · DOMESTIC

Bre-B

Launched 24/7 interoperable payments across 227 institutions in October 2025. This improves the local payment leg; it does not measure FX depth.[6]

AFRICA · REGIONAL

PAPSS

A regional payment architecture built around local-currency transactions. Morocco became its seventeenth country of presence as participation expanded.[2]

ASEAN · LOCAL CURRENCY

Local Currency Transactions

Bank Indonesia reported $14.1 billion of activity through July 2025, up 112% from the prior year. The figure measures framework activity, not institutional FX depth.[3]

SOUTH & SOUTHEAST ASIA · CONNECTIVITY

Project Nexus

Designed to connect domestic instant-payment systems across six markets. Its cross-currency design includes an explicit FX Provider role.[4]

ARAB REGION · REGIONAL

Buna

A regional multi-currency platform for cross-border payments in the Arab region.[5]

Under the hood: Pan-African Payment and Settlement System (PAPSS)

The customer-facing payment can be simple while institutional liquidity remains underneath.

Local instruction

The payer initiates through a participating institution.

Regional routing

PAPSS carries the payment across the participating network.

Liquidity is managed

The architecture can require prefunding before obligations are settled.

Net settlement

Net positions can settle through participating central-bank RTGS arrangements.[2]

PAPSS illustrates something important for our later argument. Liquidity in the narrow sense is only one part of what we refer to as deep or high-quality liquidity, which reflects the operational and infrastructure functions required to actually move money from one place to another. PAPSS needs to solve for the availability of currency. It also needs the routing and settlement work required to move it.[2]

Project Nexus exposes the same principle from another angle. Its design includes an explicit FX Provider role for cross-currency payments. Better connectivity changes how a transaction reaches liquidity. The FX provider still has to source the currencies.[4]

In both cases, the solution required operational infrastructure that exists well outside of the FX market itself.

And that infrastructure is starting to change, moving away from correspondent banks to the blockchain.

I.09

How Stablecoins Change Cross-Border FX

Stablecoins are digital tokens designed to maintain a stable value against a reference asset, most commonly the US dollar. They live on blockchain ledgers.[1]

To understand how they affect the FX market, we can start by observing how correspondent banking actually works. Banks maintain independent transaction ledgers. When a cross-border payment passes through

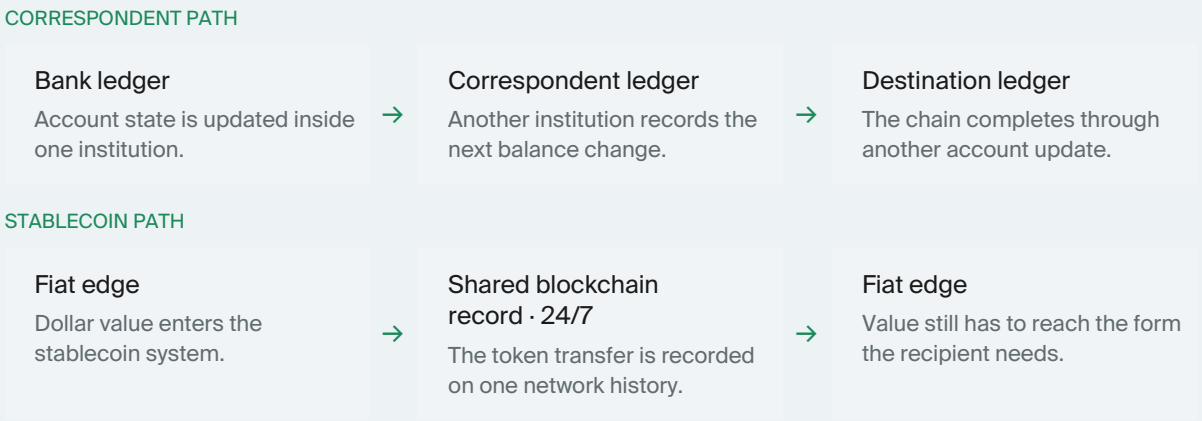
correspondent institutions, no physical currency travels from one country to another. Each institution updates its ledger to reflect the payment.[2]

Those updates have to line up across the chain. Coordinating them requires messages between institutions that still largely maintain banking hours. A payment cannot fully settle until the ledgers have been reconciled.[2]

The blockchain changes that middle step. For an on-chain stablecoin transfer, a single shared ledger records the token movement. The shared record eliminates the need for correspondent-level reconciliation between separate institutional ledgers at that step.[3]

The rail also does not close when the banks do.[4]

A different middle



This created a new way to move money across borders, one that was genuinely faster at the transfer step.

Interestingly, the dollar did not leave with the banks. About 98% of stablecoin value is denominated in US dollars.[4] The new rail largely tokenized the same currency already sitting at the center of global FX.

The edges also remain. Source money has to become usable stablecoin value before the transfer can begin. At the other end, that value still has to reach the recipient in a usable form through on- and off-ramping.[4]

These new rails still run on a lot of the old machinery. Stablecoins relocated friction from the transfer itself to the places where token money meets the banking system.

So how large has this second market become, and where is it actually being used?

I.10

How Large Is This Second Market?

As before, it depends on what you're trying to count, and what someone is trying to sell you.

Stablecoins matter in the market, but are unusually easy to misrepresent.

As of April 2026, BIS put stablecoin market capitalization at about $315 billion. This represents the total stock of stablecoin value outstanding.[1]

If we look instead at the amount of that value that moved, the number is even larger. Stablecoins generated about $35 trillion of transaction activity in 2025.[1]

It is important to realize that the vast majority of those flows were not payments. Only about $390 billion, just over 1%, were classified as payment-related.[1]

Even that is not a clean measure of cross-border payments. FXC Intelligence and Allium estimate $135 billion in non-wholesale cross-border stablecoin payments in 2025, or about 0.31% of the $44.3 trillion market they measure.[2]

All of these numbers can be true in a sense. None of them offers a complete understanding of the market it attempts to describe.

EXHIBIT A9

Stablecoin Scale Depends on What You Measure

$315B

Outstanding value

Stablecoin market capitalization, early April 2026.[1]

$35T

Raw activity

All stablecoin transaction activity during 2025.[1]

$390B

Payment-related flow

BIS estimate for 2025. ≈1.1% of raw activity 1

$135B

Non-wholesale cross-border payments

FXC Intelligence + Allium model, 2025. 0.31% of the measured $44.3T market[2]

Geography changes with the lens

Broad modeled international flows, IMF 2024

$633B

North America

$519B

Asia and Pacific

Top FXC/Allium payment corridors, 2025

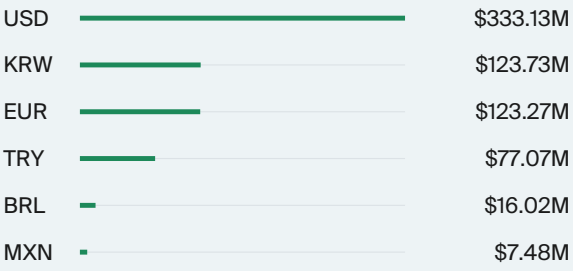
Taiwan → Turkey Taiwan → Indonesia Turkey → Indonesia
Together ≈13% of the model's stablecoin cross-border payment volume.[2]

Broad IMF geography uses a different methodology.[3]

A parallel FX market

27 fiat currencies

IMF dataset across 64 centralized exchanges. Selected Q2 2025 average daily USDT/local-fiat volume.[4]



Exchange volume shows activity, not ticket size or executable institutional depth.

The geography of penetration shifts along with the definition of flows. The IMF's broad 2024 model puts the largest absolute stablecoin activity in North America and Asia Pacific, while the FXC/Allium payment model produces a different map.[23]

Are you measuring payments? Demand for digital dollars in general? Despite its transparency, the blockchain cannot easily tell us the purpose of each transaction.

An IMF working paper observes four different USD-pegged stablecoins trading against 27 fiat currencies across 64 centralized exchanges between 2021 and November 2025.[4]

Using this model, we can see local fiat being exchanged directly for dollar-linked stablecoins.

The volume they describe is uneven. The same IMF dataset shows deep activity in some fiat pairs and very little in others. Aggregate exchange volume tells us the market exists, but still cannot tell us how the market is structured geographically or by clip size.

Liquidity becomes difficult to suss out because we cannot be entirely certain how this market reacts at scale across different locales.

So yes, this second market is real and large enough to matter, but its existing payments footprint is much smaller and harder to read than its loudest headline suggests.

The fact that we are even discussing stablecoins as legitimate payment rails reflects a recent shift. For years, much of the regulated financial system treated crypto-linked activity with caution.

Over the last 24 months, that posture has changed materially as governments have begun building regulatory structures that recognize stablecoins as financial instruments and payment infrastructure.

The question now is how far that shift has gone.

I.11

Stablecoins Enter the Financial System

Stablecoins spent years sitting awkwardly beside the regulated financial system. In the last few years, however, governments across the globe have been writing dedicated rules describing how they can be used.

The breadth of these regulations is wide, with different countries applying different tests for issuers and payment companies. These differences often reflect the particular risks that these countries have identified in the market.

Singapore draws an early line

Singapore was one of the early movers. In August 2023, the Monetary Authority of Singapore finalized a framework for single-currency stablecoins pegged to either the Singapore dollar or a G10 currency when issued in Singapore. The framework established a fairly strict regulatory regime for issuers.[1]

Europe turns the idea into a continental regime

Europe widened the perimeter. MiCA's rules for asset-referenced tokens and e-money came into full effect on December 30th, 2024. They made stablecoins an official, legal category across the European Union.[2]

The UAE treats payment tokens as a supervised payment activity

In late August of 2024, the Central Bank of the UAE's Payment Token Services Regulation took effect. It put issuance and conversion inside a Central Bank licensing framework, along with custody and transfer services.[3]

The dollar's home market moves

The United States followed with the GENIUS Act in July of 2025. It created a federal framework for payment stablecoin issuance. Full implementation began in 2026, but discussion around the bill is still ongoing. As of August of this year, a recent proposal set an expected effective date of January 18, 2027.[4]

Hong Kong moves from law to licenses

Hong Kong's Stablecoins Ordinance took effect on August 1, 2025. On April 10, 2026, the HKMA granted the first two stablecoin issuer licenses.[5]

Brazil pulls virtual assets into FX regulation

Brazil has taken another route. In November 2025, the Banco Central do Brasil adopted rules for digital asset service providers, specifying that some of their activity should be treated under the same regulatory standards as other FX conversions rather than under a new set of rules.[6]

2023 → 2026

Governments are beginning to recognize stablecoins as real payment infrastructure.

In the last three years, almost every region has laid out a framework for fiat-backed digital currencies.

EXHIBIT A10

The Regulatory Wave, 2023–2026

15 AUG 2023 — **Singapore**

MAS finalizes its framework for Singapore-issued single-currency stablecoins pegged to SGD or a G10 currency.[1]

30 JUN 2024 — **European Union**

MiCA's stablecoin titles begin applying. The broader regulation follows on 30 December 2024.[2]

31 AUG 2024 — **United Arab Emirates**

CBUAE brings payment-token services inside a licensing and registration regime.[3]

18 JUL 2025 — **United States**

The GENIUS Act becomes law. Treasury rulemaking continues through 2026 ahead of the expected 2027 effective date.[4]

1 AUG 2025 → 10 APR 2026 — **Hong Kong**

The Stablecoins Ordinance takes effect, followed by the first two issuer licenses.[5]

10 NOV 2025 — **Brazil**

BCB regulates virtual-asset service providers and brings specified virtual-asset activity into the FX framework.[6]

16 JUL 2026 — **FATF Travel Rule**

FATF reports that 83% of surveyed jurisdictions have passed legislation implementing the Travel Rule, up from 73% in 2025.[6]

Status through 2 September 2026.

By July 2026, the regulatory shift had reached the transaction itself. FATF's Travel Rule is designed to track stablecoin payment provider's transfer details, requiring them to collect and maintain appropriate records. Details like the originator and beneficiary's names and wallet numbers must "travel" with the transaction. The goal is to increase market transparency and reduce crime.[6]

Across all of these regulatory shifts, the primary goal has been legibility, making these transactions less like crypto and more like traditional payments. Banks and payment providers now have clearer answers about which stablecoins may be issued and which entities may handle them.

While there is significantly more clarity now than in years past, it's important to remember that the systems in place remain fragmentary. A stablecoin transaction might pass through several different regulatory regimes from source to recipient, and must manage with the expectations of each.

Failure means delays, a nominally highly "liquid" transaction still taking days to settle. No matter which route you follow, compliance burden will always have a role in whether a given corridor is liquid enough to complete transactions at scale.

So how do stablecoin rails stack up against the traditional market? We think we have enough context to look at that more closely.

EXHIBIT A11 · FIRST STATE

Same Economic Transaction. Different Operating Architecture.

TRADITIONAL

Source funds Sender has usable fiat value.	→	**Banking / FX / settlement** Value crosses the traditional financial stack.
→		**Recipient funds** Usable value reaches the destination.

STABLECOIN

Source funds Sender begins with usable fiat value.	→	**Fiat edge / blockchain / fiat edge** Part of the transfer path moves onto a shared ledger.
→		**Recipient funds** Usable value reaches the destination.

At this level, the two systems look more or less the same, and that is one of the key points. Both still have to turn source funds into usable destination funds while navigating the institutions and rules around them. The rail can change dramatically without making the surrounding liquidity problem disappear.

I.13

High-Quality Liquidity Extends Beyond the Rail

We have returned several times to the idea that currency availability and high-quality liquidity are different things. A market can have the asset and still struggle to convert or deliver it reliably.

If anything, the rise of stablecoins only makes this case more clear. Stablecoin transfers themselves are incredibly fast, often close to instant, but in practice they have not brought down settlement times to match.

Banca d'Italia gives us a rare view of that distinction in a live transaction. Researchers executed 200-USDC transfers across ten Italy-linked corridors. The token stayed the same while the local payment infrastructure changed.[1]

EXHIBIT A12

Fast Transfer, Uneven Outcomes

BANCA D'ITALIA · 200 USDC

The blockchain was rarely the slow part

Ten Italy-linked corridors, March 2026.[1]

Italy ↔ Brazil / Argentina

Observed end-to-end completion <20 min

Italy ↔ South Africa

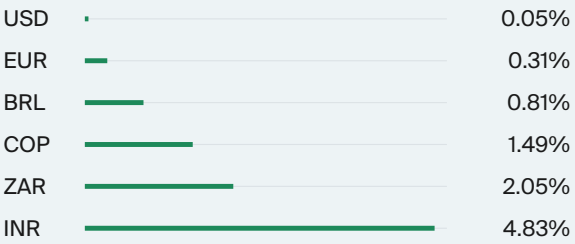
Standard bank-transfer stages remained in the path 1–2 business days

On-chain stage: generally under 15 minutes in completed routes. South Africa → Italy was around 30 minutes.

All-in pathway cost across the experiment ranged from 0.30% to nearly 9% of the amount sent.

Local price alignment varies

Mean USDT parity deviation in selected fiat markets.[2]



Parity deviation compares direct local-fiat acquisition of stablecoin dollar exposure with a synthetic route through traditional USD FX.

What this experiment can establish

Observed

End-to-end timing and pathway cost for the transactions actually executed in the experiment.

Outside the experiment INSTITUTIONAL SIZE

The study did not test $5M+ quote-to-fill execution, institutional depth or rejection behavior.

A corridor can technically have access to stablecoin-driven liquidity, while the quality of that access differs materially depending on the local infrastructure surrounding it. Markets have different banking systems and regulatory bodies, different time zones and ways of managing compliance failures. They have different local rails of different quality. Whether stablecoins are a part of that system tells us surprisingly little.[2]

Then scale makes the picture even murkier. Even this example mostly tells us what transactions look at retail sizes, small dollar amounts moved through local rails. As we discussed earlier, the capacity to fill a transaction decreases as clip size increases, particularly in markets where nominal liquidity is thin.

We cannot say what this experiment might have found if the transfer size was $10M USDC rather than 200 USDC. The study did not test institutional-size execution, so it cannot establish spread, depth or completion behavior at that scale.[1]

A fast rail can only tell us so much. To understand why one payment finishes in minutes while another takes days, we have to follow where the money actually goes.

So let's follow it.

I.14

Following $50 Million Through the Correspondent Banking System

Consider a U.S. corporate treasury that needs a supplier in Colombia to receive the Colombian peso value of $50 million.

The payment starts with dollars in a U.S. bank account and ends with pesos in a Colombian bank account. No $50 million object crosses the border. The instruction does, while banks along the route change balances on accounts they already maintain.

For this teaching transaction, assume the originating bank relies on correspondent banking to reach Colombia and that the final COP transfer runs between two banks participating in Banco de la República's CUD system. That

gives us a concrete transaction to follow from the original instruction through the currency each bank can actually use.

EXHIBIT A13

How a $50M USD → COP Payment Moves Through Correspondent Banking

1. **U.S. treasury → originating bank**
   $50M starts as a USD deposit. The treasury asks its bank to pay the Colombian supplier in COP.
2. **SWIFT**
   The bank sends an ISO 20022 payment instruction over SWIFT. The instruction crosses borders; the currency remains on bank ledgers.[6]
3. **Correspondent bank(s)**
   The correspondent chain gives the originating bank access to foreign-currency accounts and the Colombian banking system. Usable currency balances must be available when the instruction is acted on.[12]
4. **Banks along the route**
   Each bank applies its compliance controls before acting. An unresolved alert pauses the instruction while the underlying balances stay where they are.[3]
5. **Executing bank / FX desk**
   The USD/COP rate is agreed. The bank commits to the COP amount required for the payment and uses existing COP liquidity or sources any shortfall.[4]
6. **Colombian paying bank → CUD → supplier bank**
   CUD debits the paying bank's Banco de la República deposit account and credits the supplier bank. The supplier bank credits the supplier.[6]

Reconciliation follows the postings. The account-servicing bank reports settled entries and the account owner matches them against its nostro record.[6]

Assumes correspondent access into Colombia and separate Colombian paying and beneficiary banks using CUD.

1 The payment begins with dollars in a U.S. bank account

The treasury starts with a dollar deposit at its bank. It enters the beneficiary details and the amount to be paid out to the Colombian supplier.

The specific timing of the debit varies by institution. At this point, no currency crosses the border. The payment instruction is passed forward through SWIFT.

2 SWIFT carries the instruction into the correspondent chain

SWIFT does not carry currency. It is the messaging network that carries the structured payment instruction used by subsequent institutions in the chain. The current cross- border payments standard on SWIFT is ISO 20022. For a customer credit transfer, the pacs.008 message now performs the role historically associated with MT103 and can carry richer structured payment data.[6]

3 Correspondent banking provides the foreign-currency reach

The originating bank passes the request into the correspondent banking network. These institutions provide access to foreign payment systems and to usable balances in the currencies needed to complete the payment. When one bank holds an account with another, the account is a nostro from the holder's perspective and a vostro from the servicing bank's perspective.[1]

If the timing of the request leaves the correspondent account short, the institution can wait for incoming payments to supply the balance or draw on intraday credit. It can also use balances available through another banking relationship.[2] A shortfall therefore costs either time or funding.

4 The banks screen the instruction as it moves

Compliance runs inside the institutions handling the payment. The originating bank applies its controls before sending the instruction onward, and correspondent or downstream banks can screen it again before acting.

On the U.S. side, the payment is subject to OFAC sanctions requirements.[6] Colombian financial institutions operate their own AML controls under the SFC's SARLAFT framework, with suspicious-transaction reporting to the UIAF.[10]

If the instruction is flagged at any point, it can move into manual review. The instruction pauses while the relevant balances remain on the banks' books.[3]

5 A bank executes USD/COP and commits to provide the pesos

Once the instruction reaches a bank able to perform the conversion, its FX desk agrees a USD/COP rate for the $50 million payment. The bank is now committed to providing a specific amount of COP against the treasury's dollars.

The dollars the treasury gives up settle through the accounts used on the USD side of the correspondent chain. The pesos used to pay the supplier come from COP liquidity already controlled by the executing bank or its Colombian banking relationship. Dealers internalize much of their customer flow, so the bank does not necessarily buy the full COP amount in the market at the instant the customer trade is executed.[4]

If the available COP position is too small, the bank has to source the shortfall from another dealer or the market before it can fund the local payment. Any market-facing hedge or liquidity trade is a separate FX transaction with its own settlement obligation.[5]

6 The Colombian bank pays the supplier through CUD

The Colombian bank making the local payment must now have enough COP in its deposit account at Banco de la República. That balance is the money used for the final bank-to- bank transfer.

The paying bank submits the transfer instruction to CUD. If its COP balance is sufficient, CUD debits the paying bank's deposit account and credits the supplier bank's account immediately on a real-time gross settlement basis.[6] The supplier bank then credits the supplier's account.

The pesos that arrive at the supplier never crossed the border. They were released from balances already inside Colombia after the cross-border instruction had reached banks able to fund and authorize the payment.

After the postings are complete, the banks reconcile their records. The institution servicing a vostro account reports the settled entries, and the account owner matches them against its own nostro record.[6] Reconciliation confirms what posted. It does not move the payment.

The same $50 million requirement can now be followed through a stablecoin architecture.

I.15

Following The Transaction Through Stablecoin Rails

Lets look at the same transaction through stablecoin rails. The treasury begins with dollars in a bank account, the supplier still needs peso. We will be using USDC for this transfer, but it would look similar if we used USDT or any other dollar-backed stablecoin.

1 Source Fiat Processing

The treasury's $50 million begins as a bank deposit. It must then be converted into USDC. This is either done through the liquidity provider's pre-existing inventory, or the stablecoins are minted for purpose.[1]

The providers have their own liquidity requirements. Regulations might require issuers to preposition fiat to support the payment.[2] An onchain asset still depends on usable bank money being available.

2 Compliance Screening

Before the USDC can move, the institutions handling the payment have to approve the transaction. The treasury and supplier have already been identified, but the payment itself may still be screened for sanctions, suspicious activity and other compliance requirements.[3]

For a $50 million transfer, those checks can involve more than simply checking two wallet addresses. The provider may need information about the sender, the beneficiary and the purpose of the payment. If required information is missing, the transfer stops here until it is resolved.

Once those checks are complete, the provider can release the USDC for transfer.

3 Transfer

The $50 million equivalent in USDC now moves from the sending wallet to the receiving wallet onchain.

This is the part stablecoins change most dramatically. The transfer can settle in seconds or minutes, depending on the network being used, and it does not depend on banking hours.[4]

But the supplier still does not have pesos.

4 FX Conversion / Off-Ramp

The USDC now has to be sold for Colombian pesos.

For a small payment, that conversion might be relatively easy. At $50 million, the question is whether enough COP liquidity is actually available at the quoted price. A provider may source that liquidity from its own inventory or from counterparties in the local market.[5]

This is where the size of the payment matters again. Blockchain settlement tells us that the USDC arrived. It tells us nothing about how much COP can be bought, at what spread, or how the price changes as the order gets larger.

Once the conversion is complete, the provider has COP available for delivery.

5 Destination Delivery

The pesos still have to reach the supplier's bank account.

The provider or its Colombian banking partner sends the COP through the domestic banking system. For a payment of this size, that may involve Colombia's large-value settlement infrastructure, including CUD.[7]

The payment can only move this way because the provider and payout bank were already connected to the fiat and stablecoin systems before the $50 million instruction arrived.

What had to exist before the payment arrived

Banking Relationships give the provider somewhere to receive the treasury's dollars and give the payout bank access to COP. On the U.S. side, the provider needs a bank account connected to USDC issuance or inventory.[1] In Colombia, the payout bank needs access to domestic settlement through CUD.[7]

Regulatory Licenses / Market Access allow the U.S. provider to handle the stablecoin leg and the Colombian payout bank to make the local payment. Those permissions are already in place when the $50 million instruction arrives.

Liquidity Pools are the balances that make the payment possible. USDC must be available at source, while COP must be available in Colombia. The provider can hold the USDC while the payout bank holds the pesos; current provider arrangements use this split, with the stablecoin provider advancing USDC and the payout partner paying from its local treasury.[8] A large issuer reserve cannot substitute for COP sitting with the payout side.

Compliance Infrastructure supplies the customer records and screening systems used when the $50 million payment reaches Compliance Screening.

Technical Infrastructure supplies the wallet access and blockchain connectivity needed to move and monitor the USDC. It does nothing to create COP for the final payout.

I.16

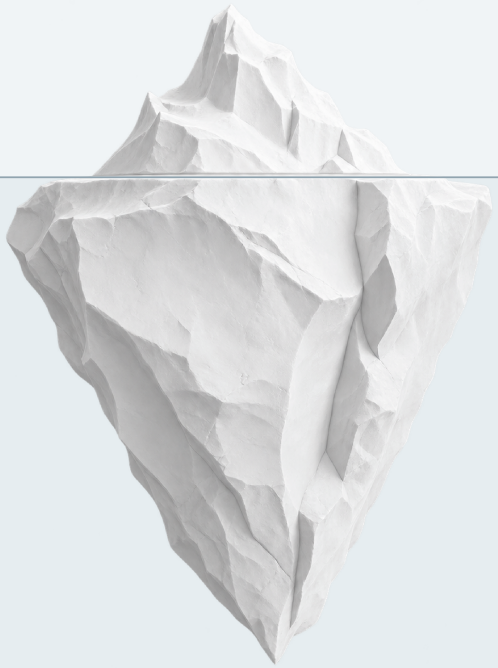
Two Architectures, One Transaction

The transaction we just followed is what we call the FX Iceberg.

It separates the payment the customer sees from the Operations required to complete it. The Infrastructure sits around that process, supplying the access and capacity each Operation depends on.[1]

EXHIBIT A14

The FX Iceberg



CUSTOMER VIEW

$50M USD → COP

WATERLINE

5 **Destination Delivery**
COP reaches the supplier bank

4 **FX Conversion / Off-Ramp**
USDC finds executable COP liquidity

3 **Transfer**
USDC moves between wallets

2 **Compliance Screening**
Controls determine whether the payment can advance

1 **Source Fiat Processing**
USD funding becomes usable stablecoin-side value

Infrastructure — Persistent capacity around the transaction

Banking Relationships	Regulatory Licenses / Market Access	Liquidity Pools	Compliance Infrastructure	Technical Infrastructure
Fiat access at both edges	Standing permission to operate	Usable capital in the right form and place	Identity data and monitoring	Wallet and chain connectivity

Now put the two $50 million transactions side by side.

Transfer is the clearest relocation. In correspondent banking, the instruction and ledger updates move across institutions. In the stablecoin path, USDC moves onchain and the blockchain provides a common record for that leg. Reconciliation at the transfer layer changes with it; source funding and the later fiat legs still sit outside that shared record.[2]

Working capital moves as well. The traditional path can draw on bank balances or credit.[3] The stablecoin path can draw on USDC inventory or credit, while COP still has to be available in Colombia for conversion and payout.[4]

EXHIBIT A11 · SECOND STATE

Where the Function Went

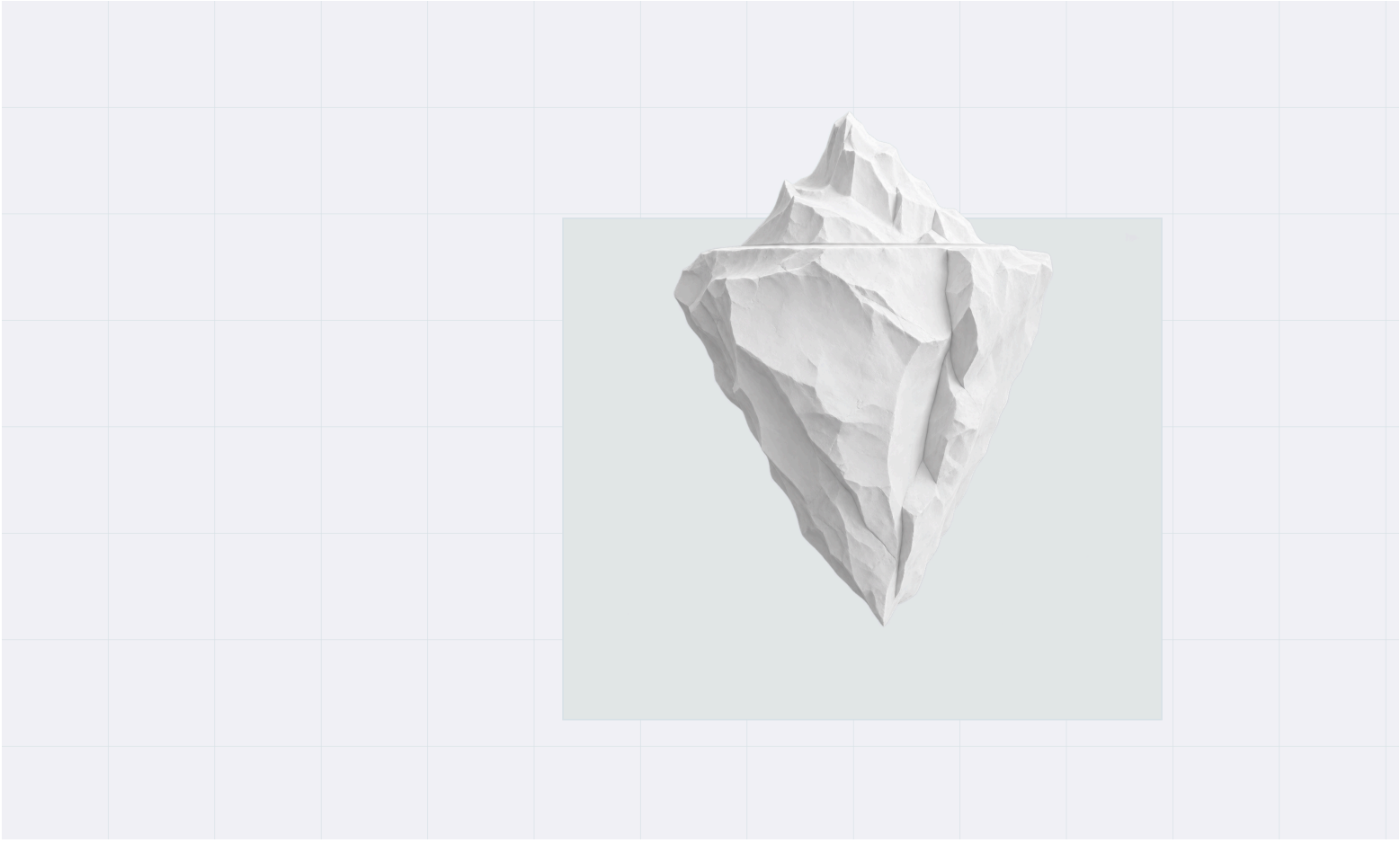
TRADITIONAL	STABLECOIN
Source funds USD in the treasury's bank account.	**Source funds** USD in the treasury's bank account.
↓	↓
Transfer Bank instructions move across institutions.	**Transfer** USDC moves onchain.
Ledger record Separate bank records are updated and reconciled.	**Ledger record** The blockchain provides the common record for the token leg.
Working capital Bank balances or credit support payment and settlement.	**Working capital** USDC inventory or credit can support the sender side.
FX execution USD is sold for COP.	**FX execution** USDC is sold for COP.
Settlement Bank and FX obligations settle through financial-market infrastructure.	**Settlement** USDC settles onchain; the fiat edges still settle through banks.
Banking access Correspondent and local-bank relationships provide fiat reach.	**Banking access** Fiat banks remain necessary at source and destination.
Regulatory path Banking, payment and FX permissions govern the path.	**Regulatory path** Stablecoin activity sits alongside payment and FX permissions.
Destination delivery COP reaches the supplier through the Colombian banking system.	**Destination delivery** COP reaches the supplier through the Colombian banking system.
↓	↓
Recipient funds Usable COP in the supplier's bank account.	**Recipient funds** Usable COP in the supplier's bank account.

Before we can compare liquidity across the two systems, we need to be precise about what we are measuring.

ACT I | 01

THE CURRENT STATE OF FX

Stablecoins reorganize the FX Iceberg.



Stablecoins aren't sufficient to guarantee a reliable, fast FX transfer. They do not eliminate the problems of traditional FX, they relocate them.[56]

How To Measure Liquidity Quality

02



II.01

Measuring High-Quality Liquidity

If liquidity determines whether a cross-border payment actually works, then measuring liquidity is one of the most important things you can learn to do.

The market is full of numbers that appear to answer that question:

Trading volume measures activity.

Stablecoin reserves describe backing.

A quoted spread says something about price.

24/7 transfer availability tells you when the rail is available.

Each tells us something useful. None tells us whether a provider can move the amount you need, at the price you expect, and deliver usable money reliably.

That is the problem we want to solve in this Act. We want to give you a framework you can use to compare different industry claims and determine whether the liquidity you're being offered is truly “high quality.”

The first step is to be precise about what each claim is actually measuring.

EXHIBIT B1 · FIRST STATE

What Is Being Measured?

PROVIDER CLAIM	WHAT IT MEASURES	WHAT REMAINS UNKNOWN
“Our spreads are tiny.”	Quoted or observed spread at a stated size and moment	Will your full order execute near that price?
“Our rails operate 24/7.”	Transfer-layer availability	Can the complete payment finish outside local banking windows?
“We support this country.”	Nominal geographic reach	What size can be converted and delivered there reliably?
“We have deep liquidity.”	Undefined until executable size and market conditions are specified	How much can actually be filled before price or availability deteriorates?
“Our transactions settle in seconds.”	Undefined until “settlement” is defined	Does this mean the token transfer settled, or that usable money reached the recipient?

High-quality liquidity has to be measured through the transaction itself. The question is how well that liquidity performs when someone actually tries to use it.

That is what our Liquidity Scorecard is designed to measure.

II.02

The Liquidity Scorecard

A useful liquidity framework has to catch more than one kind of failure. A tight quote can deteriorate as transaction size rises. A trade can execute successfully and still take too long to reach the beneficiary.

We use six measurements to look at the transaction from quote through delivery.[1] Each answers a different question about whether the liquidity being offered remains usable in practice.

EXHIBIT B3

The Liquidity Scorecard

METRIC	QUESTION	WHAT TO OBSERVE	DETERIORATION LOOKS LIKE
Spread Consistency	Does the price being offered remain usable?	Spread across time, transaction size and changing market conditions	Spreads widen materially as size rises or conditions become more difficult
Execution Quality	Do you get the trade you expected?	Fill, slippage, partial fills and realized market impact	The realized price or amount deteriorates from the quote
Rejection Rate / Last-Look Frequency	Does apparently available liquidity disappear when you try to use it?	Rejections, declines, expiries and their behavior across size or stress	Rejections become more frequent when the transaction becomes harder to execute
Settlement Speed Under Load	Does the complete payment still finish reliably?	End-to-end completion time and failures as size or system load changes	Completion slows or becomes less reliable as demands on the system rise
Depth-of-Book Consistency	Is meaningful capacity available behind the first quote?	Executable depth, replenishment and performance across repeated flow	Larger orders consume liquidity quickly or available capacity fails to replenish
Price Accuracy / Alignment	Does the offered price track the relevant market benchmark?	Deviation from an appropriate benchmark across time and conditions	Prices diverge persistently or increasingly from the benchmark

The Scorecard gives us six places to look when a liquidity claim is being tested. We will start with the first two: the price you are shown, and the trade you actually get.

II.03

From Quoted Price to Realized Trade

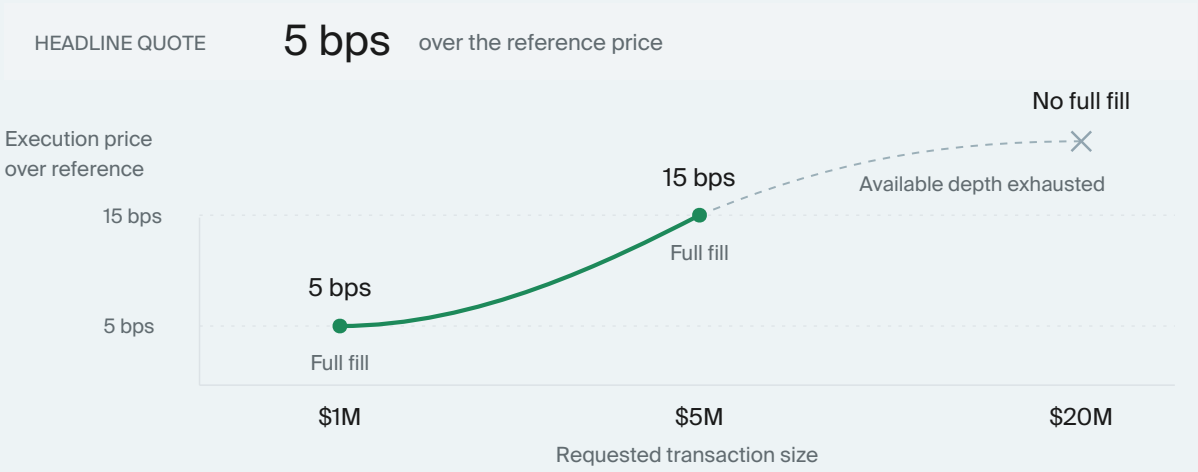
A quoted price is only evidence of liquidity if you can actually trade with it.

Spread Consistency is the first measure of “tradeability,” it asks whether the price being offered remains stable across the various conditions you will be trading under. A provider might technically offer 3bps, but only for a scale and timing of trades you don't actually need.

Execution Quality comes into play after you accept the quote. It measures what actually fills: whether your full order is deliverable, and how likely it is for your trade to fail in transit.

EXHIBIT B3

The headline quote isn't the entire story.



Earlier, we saw that actual book depth could also vary widely depending on clip size. For some currency pairs, the historical order books did not contain enough liquidity to fill the largest transaction sizes at all.[1]

The same problem remains relevant today, whether we are talking about traditional FX or stablecoin- driven rails. Larger spot trades still influence execution method because clients and dealers have to manage market impact.[2] A provider should therefore be evaluated at the ticket size and execution method you actually plan to use.

The goal here is reliability, not whether someone can quote you a price, but how likely it is for you to be able to transact at that price over time.

II.04

From Accepted Request to Completed Settlement

Reliability in cross-border FX is a two-step process. The first, can you actually access the price you were shown? The second, once that trade executes, how predictably does the transaction reach the account you requested.

Last look is emblematic of the first step. Generally, accepting a quote does not guarantee execution. The liquidity provider retains a short window in which they can choose to decline the trade. Under the FX Global Code, providers using last look are expected to disclose how the process works.[1]

For some providers, these types of rejections are relatively rare and non-disruptive, but for others the rejection rates can become large enough to affect execution quality. A 2022 review by Hong Kong's Securities and Futures

Commission found last-look rejection rates between 1.48% and 2.80% among the selected firms it examined.[2] If you are comparing providers, it is useful to look at their fill rate for the specific corridor and the market conditions you actually trade under.

EXHIBIT B4

Reliability is a two-step test.

STEP 1 · ACCESS

Can you actually get the trade?

Price shown → You request → Accept / Reject

1.48–2.80%

Last-look rejection rates among selected firms reviewed by the Hong Kong SFC.

Historical institutional observation · review published January 2022

WHAT TO ASK

How often do requests fill in the corridor you use?

How long does a rejection take when markets are moving?

STEP 2 · COMPLETION

How predictably does the money arrive?

TRADE EXECUTED — USABLE MONEY

Redemption processing took longer than issuance on average.

Federal Reserve Bank of New York matched stablecoin primary-market activity to interbank payments.

After March 2023, the redemption-delay gap widened.

The authors interpret the divergence as evidence of greater liquidity-management needs.

Stage-level processing evidence, not end-to-end cross- border completion.

WHAT TO ASK

How long do payments normally take to complete?

What happens to the tail when volume or liquidity pressure rises?

Last Look is one half of reliability. The other half is settlement speed under load.

A trade can execute at exactly the price you expected and still produce a poor outcome if the transaction then takes longer than expected to complete. Once the trade is accepted, the question changes. How long does it take for usable money to actually reach the account you requested?

Settlement Speed Under Load measures how predictable that process is. A provider that usually settles quickly may behave very differently when transaction sizes increase, volumes spike, or the market is under stress. What matters is whether settlement remains consistent under the conditions in which you actually need it.

Settlement speed is partly a liquidity-management problem. A fast transfer mechanism does not guarantee that the money around it will always be available just as quickly. Federal Reserve Bank of New York research found that stablecoin redemptions took longer to process than issuances on average in its matched sample. After March 2023, that gap widened further as liquidity-management needs increased.[3] The study looks at stablecoin primary-market processing rather than an entire cross-border payment, but the mechanism is the important part: when sourcing or releasing liquidity becomes harder, completion time can stretch with it.

That delay matters because the underlying payment obligation does not wait for the FX market. If you owe a supplier COP 200 billion and the conversion is delayed, a move in USD/COP can change how many dollars you need to deliver the same COP amount. The invoice is still COP 200 billion. What changed is the cost of paying it. If the FX rate was already locked but the trade fails to settle and has to be replaced, you may still have to return to the market at a worse price while the original payment remains due. The longer the transaction

remains unresolved, the more opportunity there is for the economic outcome you expected to diverge from the one you ultimately get.

That is why settlement speed should be evaluated the same way as execution quality: not by the provider's best-case claim, but by what happens across real transactions. How long do payments normally take? What does the tail look like? What happens when the system is busy? And how often does a delay become material enough to change the economic outcome?

Even reliable execution and settlement do not tell you whether the same capacity will still be there for the next large trade.

II.05

Persistent Capacity and Price Alignment

One good trade does not tell you how much liquidity will be there for the next one.

Depth-of-Book Consistency asks whether liquidity comes back after you use it. If you execute a $5 million trade, can you execute another $5 million trade an hour later at a similar price? If the second trade is materially worse or cannot fill, the provider had enough depth for the first transaction, but not enough to support the flow you actually need.

There is another complication. Several visible quotes can come from the same underlying liquidity. In fragmented FX markets, the same trading interest can appear on multiple venues and disappear elsewhere once it trades in one

place.[1] So four prices on a screen do not necessarily mean four independent sources of liquidity.

EXHIBIT B5

Can you do it again, and at what price?

DEPTH-OF-BOOK CONSISTENCY

How much independent depth is actually there?

Venue A	Venue B	Venue C	Venue D
Quote available	Quote available	Quote available	Quote available

can reflect the same ↓

Underlying trading interest

Interest trades on one venue →

Other displayed quotes disappear

DECISION TEST

What happens when you need to execute the same size again?

PRICE ACCURACY / ALIGNMENT

Does another route give you the same dollar exposure more cheaply?

Direct route

Local fiat → USD stablecoin

Synthetic route

Local fiat → USD → USD stablecoin

Parity deviation

The difference between the direct stablecoin price and the synthetic route through traditional USD spot FX.

DECISION TEST

How does the price you can trade compare with an alternative route?

For a provider, repeated execution is more useful than a snapshot. Trade the size you need, then see whether they can do it again.

That still leaves price. A provider can fill your trades reliably and still quote you poorly relative to the rest of the market.

Price Accuracy / Alignment asks how the price you can trade compares with a relevant alternative. Stablecoin FX gives us a useful example. If you want dollar stablecoin exposure using COP, you can buy the stablecoin directly with pesos. You can also convert COP into USD through the traditional FX market and use those dollars to buy the same stablecoin.

The IMF measures the difference between those two routes as parity deviation.[2] It is a measure of price alignment, not the bid-ask spread or the total cost of making the payment.

That difference also changes with demand. In the IMF study, a 1% exogenous increase in net stablecoin inflows increased parity deviation by about 40 basis points.[2] As stablecoin demand rises, the direct route can become more expensive relative to the traditional FX route.

So far, we have mostly asked whether liquidity holds under ordinary conditions. The harder test is what happens when size rises or the market turns volatile.

II.06

What Changes When Conditions Change?

So far, we have treated the six measurements separately. That was useful for learning them, but liquidity rarely behaves one metric at a time.

A provider can show you a tight spread while offering very little depth behind it. A trade can fill cleanly and still take too long to reach the account you requested. The Scorecard becomes most useful when you read the measurements together.

EXHIBIT B6

One strong reading can hide weakness somewhere else.

STRONG READING	WHAT IT CAN MISS	READ TOGETHER
Spread Consistency Tight quoted price The price looks competitive at the size being shown.	Depth Consistency Shallow capacity Larger or repeated trades quickly consume what is available.	The quote works until you ask it to carry more size. A good spread alone does not establish the capacity you actually need.
Execution Quality Clean fill The requested trade executes as expected.	Settlement Speed Under Load Slow completion The money still takes too long to reach its destination.	The trade succeeds while the payment performs poorly. Execution quality cannot stand in for end-to-end reliability.
Depth Consistency Repeatable capacity The provider can keep filling the transaction size you need.	Price Accuracy / Alignment Poor market alignment The executable price remains worse than a relevant alternative.	Plenty of capacity can still be expensive capacity. Reliable execution does not tell you whether the economics are competitive.

We have already seen one version of this with transaction size. As larger orders consumed more of the available book, execution became more expensive and in some markets the available depth itself became the constraint. The point here is the intersection: the headline price could still look attractive even while the amount you could actually trade at comparable terms was deteriorating.

Market stress can produce different combinations again. On 2 April 2025, U.S. tariff announcements surprised investors and were followed by a sharp increase in FX volatility, a weaker dollar, and a surge in trading as market participants adjusted their currency exposure.[1] Yet major-market liquidity held up comparatively well. BIS analysis found no clear signs of market dysfunction, bid-ask spreads remained relatively resilient, and price impact stayed well below the levels seen during the March 2020 shock.[1]

The Swiss-franc shock on 15 January 2015 produced a very different pattern. After the Swiss National Bank unexpectedly removed the EUR/CHF floor, the franc was roughly 11% stronger after two minutes and roughly 19% stronger after seven. Market illiquidity rose sharply, bid-ask spreads widened, and average forward trade size fell by about 34%.[2]

Both episodes put pressure on the market. The Scorecard matters because pressure did not produce the same result. A label such as “volatile” or “stressed” tells you about the environment. The measurements tell you what happened to the liquidity.

That creates a practical problem. Reading the intersections requires evidence across more than one metric at the same scale. Public data do not always give you that.

II.07

What You Have to Ask Your Liquidity Provider

The Scorecard creates a second problem: most of the provider-level data you would want to evaluate is not public.

That is partly a feature of the market itself. FX is decentralized and fragmented, and much dealer- customer trading happens away from public order books. Provider-level measures such as fill behavior or settlement tails are generally not published as standardized market statistics.[1]

In newer stablecoin markets, the problem is sharper. In the public sources we reviewed, we could not find a neutral institutional-scale local-fiat dataset that would let us populate the Scorecard end to end. Much of the useful execution telemetry sits inside private liquidity providers.

That leaves buyers with the claims providers choose to make. “Tight spreads.” “Deep liquidity.” “Fast settlement.” Those claims may be accurate.

We do not expect that information asymmetry to disappear soon.

CLAIM → QUESTION

What can change is the quality of the questions you ask.

A best-case number can still tell you very little about how performance changes when size rises or conditions become harder.

EXHIBIT B7

Turn the headline claim into a question you can evaluate.

METRIC	HEADLINE CLAIM	BETTER QUESTION
Spread Consistency	“Our spreads are tight.”	Show me the spread distribution at the ticket sizes I trade. How does it change when volatility rises?
Execution Quality	“We have a high fill rate.”	What fill rate and slippage do you see at my transaction size? What do the worst outcomes look like?
Rejection / Last Look	“Execution is reliable.”	How often are requests rejected, and how does that change when size or volatility rises?
Settlement Speed Under Load	“Settlement is fast.”	Show me typical completion time and the slow tail by transaction size. What happens during high-volume periods?
Depth-of-Book Consistency	“We have deep liquidity.”	How much can I execute at comparable terms, and how quickly does capacity return after a large trade?
Price Accuracy / Alignment	“Our pricing is competitive.”	How does my realized price compare with the relevant benchmark or alternative route over time?

The difference is practical. Asking for a spread invites a snapshot. Asking how that spread behaves at your size and during the conditions you actually trade turns the claim into something you can test.[2]

The evidence should match the claim. A description of RFQ capability cannot tell you the fill or slippage you actually receive. A best-case settlement promise cannot tell you what happens in the tail.

A provider that can answer these questions with transaction-level evidence gives you something you can evaluate. If the answer keeps returning to the headline claim, the Scorecard has still done useful work: it has shown you what remains unknown.

Public evidence still matters. It gives you independent context and helps test whether the mechanisms behind a claim are plausible. It usually cannot tell you how a specific provider will perform for your transaction.

II.08

The Other Side of the Scorecard

Up to now, we have mostly been talking to the people buying liquidity: treasurers and product teams trying to decide whether a provider can actually move the money they need.

If that is your job, the Scorecard gives you a practical place to stop. It gives you a way to test the claim rather than simply accept the headline.

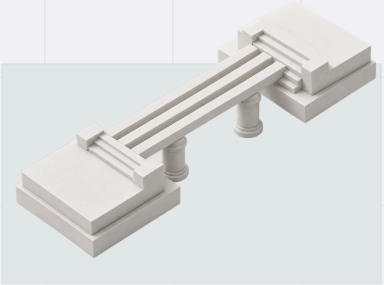
But some of you are on the other side of the conversation.

If you are building a stablecoin-driven payments product or trying to become the liquidity provider yourself, unfortunately, things are about to get more complicated.

Offering high-quality liquidity means being able to reproduce the outcomes the Scorecard measures. The quoted price has to be executable, and an accepted trade still has to reach the destination as usable money. That performance depends on what sits underneath.

We think the answer sits at the intersection between Operations and Infrastructure. To see what that takes, and what happens when one of those dependencies fails, we need to return to the FX Iceberg.

What High-Quality Liquidity Requires



03

III.01

Back to the FX Iceberg

Earlier, in section one, we introduced the idea of the FX Iceberg.[1] This framework was organized around two major parts: five Operations that formed the body of the iceberg, and the water surrounding that body, representing the Infrastructure that supports those Operations.

We chose this specific structure for a reason: interdependence.

An iceberg and the water surrounding it are intrinsically linked, parts of the same physical system. Change the conditions of one, and the other will be affected. Cross-border payments work the same way. The condition of the Infrastructure changes what the Operations are capable of doing, and different use cases require different Infrastructure configurations around those Operations.

A remittance provider in the UAE can require a different organization of Operations and Infrastructure than a fintech in Hong Kong or a payroll company in Mexico. The basic shape remains the same, while the interactions between the two can drive wildly varying outcomes.

Consider the off-ramp. COP cannot reach the supplier's account unless the provider has a working relationship with a local banking or payout institution capable of delivering it. The provider also needs the market access and permissions required to operate there. Even the blockchain Transfer depends

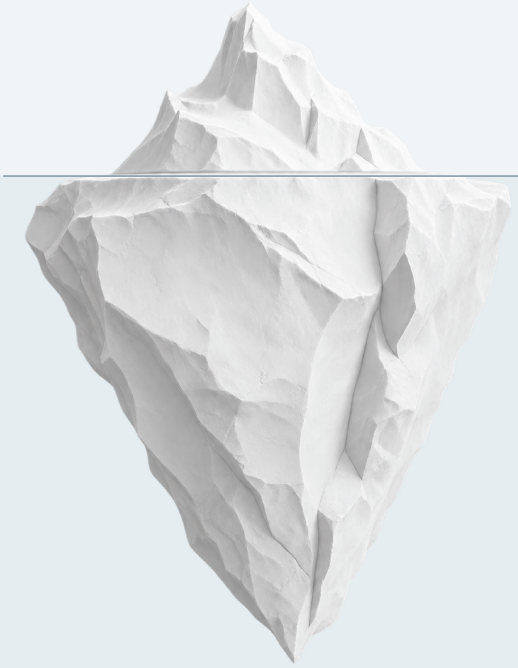
on Technical Infrastructure that can authorize and orchestrate the movement once stablecoin liquidity is available.

High-quality liquidity depends on the interaction between operational capacity and the Infrastructure supporting it. A weakness in either can stretch a transaction from minutes to hours or days, even while the stablecoin Transfer itself remains nearly instantaneous.

Over the next few sections, we follow the five Operations again and look at how those relationships work in practice.

EXHIBIT C1 · FIRST STATE

The FX Iceberg is an interdependent system



CUSTOMER VIEW

$50M USD → COP

WATERLINE

5 **Destination Delivery**
COP reaches the supplier bank

4 **FX Conversion / Off-Ramp**
USDC finds executable COP liquidity

3 **Transfer**
USDC moves between wallets

2 **Compliance Screening**
Controls determine whether the payment can advance

1 **Source Fiat Processing**
USD funding becomes usable stablecoin-side value

INFRASTRUCTURE — Persistent capacity around the transaction

Banking Relationships	Regulatory Licenses / Market Access	Liquidity Pools	Compliance Infrastructure	Technical Infrastructure
Fiat access at both edges	Standing permission to operate	Usable capital in the right form and place	Identity data and monitoring	Wallet and chain connectivity

Let's start where the treasury starts: the USD has been sent. When can the provider actually use it?

III.02

Source Fiat Processing

The treasurer has sent $50M USD. Now the liquidity provider has to find enough USDC to fund the payment.

The provider can find these stablecoins in a number of ways. Some keep a pre-positioned inventory of USDC in their own wallets and allocate it as necessary.[2] If more USDC is needed than they can support natively, the provider has to source it from an issuer like Circle. In those cases, they provide the issuer with fiat from a linked bank account, and the issuer provides them with newly minted stablecoins.[1]

To make that handoff reliable, the liquidity provider needs a banking relationship that can receive the $50M in fiat, and wallets on the appropriate networks for the stablecoins. If either doesn't exist, the payment cannot go anywhere.[2] Even in a better case, if the payment provider doesn't have a clear means of funding transactions at the clip sizes they work with, delays should be expected.

This is where stablecoin orchestration comes into play. Orchestration is the provider's ability to find the stablecoins it needs to transact across disparate networks. After they have drawn down their own inventory, the orchestration layer needs to know where else the coins can be sourced and how to get them into the payment flow. Under ordinary conditions, that might just mean sending

dollars to Circle and minting the difference. Other conditions can make that route insufficient.[2]

The collapse of Silicon Valley Bank in March 2023 highlights the kinds of problems that can arise when liquidity sources are too limited. SVB failed after a rapid run on deposits, and Circle disclosed that $3.3 billion of the reserves backing USDC were stuck at the bank. With those reserves inaccessible, Circle's primary market largely stopped operating over the weekend, while USDC fell as low as $0.86 in secondary markets.[3]

Now imagine our provider is sitting on 50M USDC to fund their $50M payment. At $0.86, the coins only represent $43M in market value. The provider is suddenly $7M short.

What are they to do?

Their usual minting source is unavailable, so the orchestrator has to find another route. One option might be another USD-backed stablecoin like USDT. In a corridor with a sufficiently liquid local- currency stablecoin, that might be another. They could also go to the public markets, though sourcing this way exposes them to volatility.[3]

As USDC moves up or down, the number of coins needed to cover the same shortfall changes with it. At transaction sizes this large, simply finding sellers can be an issue.

Ultimately, each of these failure states can slow settlement or increase cost. In the worst case, the transaction fails. De-pegging events like SVB are relatively rare. Less dramatic orchestration challenges exist across the market every day.

High-quality liquidity requires having systems in place to navigate these issues as they appear.

EXHIBIT C2 · SOURCE FIAT PROCESSING

From treasury USD to usable USDC

FX ICEBERG

5 Destination Delivery | 4 FX Conversion / Off-Ramp | 3 Transfer | 2 Compliance Screening | **1 Source Fiat Processing**

SOURCE FIAT PROCESSING

$50M USD reaches the provider	The treasury's dollars arrive through the provider's bank.
↓	
USDC inventory is allocated or a shortfall is sourced	The provider uses coins it already holds or its orchestration layer finds another source.
↓	
USDC is available for the payment	The stablecoins are ready for the next Operation.

WHERE WEAK INFRASTRUCTURE CAN INTERRUPT THE FLOW

Banking Relationships

The $50M cannot reach the provider on time or at the required size.

Liquidity Pools

USDC inventory is short, so the provider has to source more before the payment can continue.

Technical Infrastructure

The orchestration layer cannot reach another source when inventory is short or the usual minting route fails.

Stress point: a depeg can impair the value of USDC already held while primary-market minting is unavailable. The provider now has to cover the shortfall through another route or a volatile secondary market.[3]

SUPPORTING INFRASTRUCTURE

Banking Relationships | Liquidity Pools | Regulatory Licenses / Market Access | Compliance Infrastructure | Technical Infrastructure

Once the USDC is available, the question changes: can the transaction be released? That is the job of Compliance Screening.

III.03

Compliance Screening

The liquidity provider has determined how they will source the stablecoins. Before they can take the next step, they have to prepare the transaction for compliance checks on both sides.

For a transaction that originates in the United States, the provider needs to collect enough information for the payment to stand up to the checks applied by the U.S. financial system. The treasurer's bank already knows its customer, but the transfer must still pass through required BSA/AML controls. Covered transfers must also carry forward the information FinCEN requires about the sender and beneficiary. If the bank discovers a potential OFAC sanctions match, that has to be resolved before the transaction can be completed.[1]

Some of that information also has to reach Colombia. Under the FATF Travel Rule framework, required originator and beneficiary information can travel separately from the USDC as long as it reaches the receiving institution when it is needed.[2]

When the USDC arrives in Colombia, the receiving financial institution processes the transaction through its own controls before the funds can be released. Supervised Colombian financial institutions operate under SARLAFT,

which requires them to monitor money-laundering and terrorist- financing risk, investigate alerts and make the required reports to UIAF.[3]

The potential points of failure are myriad. Suppose the source-side checks are completed and the liquidity provider releases the USDC. The blockchain transfer finishes. When the Colombian institution receives the transaction, it finds that the beneficiary information is incomplete or that a name is misspelled. The coins have arrived. The payment has stopped.

The provider now has to get the missing information or whatever supporting documentation the Colombian institution needs to clear the transaction under SARLAFT. A provider with a good process can resolve that quickly. A weak process lets the delay grow. If the transaction cannot be cleared at all, the provider may have to unwind the route and return the funds.

These sorts of delays can change the economics of the payment while it is in flight. During review, the price of pesos might move. By the time the hold clears, the provider may need to convert at a different price from the one it expected when the payment began.

Operators with robust Compliance Infrastructure understand how to deal with these contingencies quickly. They know what the corridor's financial institutions need and are able to provide it reliably. The provider can collect that information early and get it to the right institution as the transaction moves. When a problem appears, it has a process for resolving it.

They also have the appropriate Regulatory Licenses / Market Access to make the transaction in the first place, and understand the reporting and documentation rules that apply in the corridor.

EXHIBIT C2 · COMPLIANCE SCREENING

Preparing the payment to clear both sides

FX ICEBERG

5 Destination Delivery | 4 FX Conversion / Off-Ramp | 3 Transfer | **2 Compliance Screening** | 1 Source Fiat Processing

HOW THE PAYMENT CLEARS

U.S. source-side review	The provider supplies the required information. The U.S. financial institution applies its BSA/AML controls and resolves any OFAC sanctions issue before the payment advances.
↓ USDC Transfer	The token moves onchain.
↓ Colombian destination review	The receiving institution applies SARLAFT controls before the payment can advance toward conversion and payout.
↓ Cleared to continue	The payment can move into FX Conversion / Off-Ramp and local delivery.

The provider prepares the information the route requires. It collects the transaction details and gets them to the institutions that need them. Under the Travel Rule framework, required originator and beneficiary information can travel outside the blockchain.[2]

WHERE WEAK INFRASTRUCTURE CAN INTERRUPT THE FLOW

U.S. source side

Required information is missing, BSA/AML review cannot be completed, or an OFAC match has to be resolved before the payment can advance.

Colombian destination side

The USDC arrives, and conversion and payout wait while the receiving institution resolves a SARLAFT alert or requests more information.

SUPPORTING INFRASTRUCTURE

Banking Relationships | Regulatory Licenses / Market Access | Liquidity Pools | Compliance Infrastructure | Technical Infrastructure

Once the source-side checks clear, the provider can release the USDC. The destination still has its own review waiting in Colombia. Between them sits the part stablecoins are famous for: Transfer.

III.04

Transfer

The source-side checks have cleared, so the provider releases the USDC.

Under ordinary conditions, this part is fairly simple. The provider signs the transaction, submits it to the network and waits for finality. Once the network finalizes it, the receiving side controls the coins.

That simplicity depends on the technical system underneath it working.

On 6 February 2024, Solana stopped finalizing blocks at about 09:53 UTC. The outage came from a software bug involving how programs were compiled and cached. Validators had to coordinate a restart, and normal consensus progress did not resume until about 14:55. For roughly five hours, a provider using that path could have the payment funded and cleared while the network itself was unable to complete the Transfer normally.[1]

This brings us back to orchestration. Earlier, the orchestration layer had to know where the provider could find the stablecoins it needed. During Transfer, that layer has another job: it needs to know whether the route is working and whether the transaction actually finalized. If the route fails, it needs to know whether another supported network can still reach the receiving side.

An alternate network only helps if the provider already supports it and the destination can receive the asset there. Otherwise, the payment waits for the network to recover.

Meanwhile, the supplier waits.

When finality returns, the Transfer can complete. With the token across, the problem changes again: the provider has to turn USDC into pesos.

EXHIBIT C3 · TRANSFER

Transfer works only while the technical path stays live

FX ICEBERG

5 Destination Delivery | 4 FX Conversion / Off-Ramp | 3 Transfer | 2 Compliance Screening | 1 Source Fiat Processing

TRANSFER

Provider signs the transaction	The sending wallet authorizes the USDC transfer.
↓ Transaction is submitted	The provider sends it to the network.
↓ Network reaches finality	The ledger confirms the transfer.
↓ Receiving side controls the USDC	The Transfer Operation is complete.

WHERE WEAK INFRASTRUCTURE CAN INTERRUPT THE FLOW

Technical Infrastructure

Transfer depends on the underlying network continuing to finalize blocks. If consensus stalls, a submitted transaction cannot reach finality and the payment cannot complete on that route.

SUPPORTING INFRASTRUCTURE

Banking Relationships | Regulatory Licenses / Market Access | Liquidity Pools | Compliance Infrastructure | Technical Infrastructure

III.05

FX Conversion / Off-Ramp

First, they'll need someone willing to quote them a price.

At this size, the provider may need to deal with several counterparties to find sufficient liquidity. They will send out requests and brokers will return with quotes. The existence of quotes does not by itself ensure the availability of fiat. Many of these quotes might actually be “mirage liquidity,” aggregated prices all

pointing to the same underlying pool of fiat. The provider will need systems to determine how much real capital exists in the corridors they operate within.[1]

Failure comes in many forms that cut across the Iceberg. Suppose the first counterparty can only cover part of the amount, and the next quote is at a price that is untenable. The provider might have to wait for more liquidity to become available, or eat the additional costs. With deeper liquidity pools, the provider would have more counterparties to search against, increasing their odds of filling at a reasonable price.

Technical infrastructure is what makes that search possible, compliance infrastructure ensures that as the newly acquired fiat makes its way into local banks it is not held up or turned away.

Colombia adds a specific market-access constraint. Banco de la República says digital assets, including stablecoins, are not treated as foreign currency under the Colombian exchange regime, the SFC has not authorized supervised financial institutions to custody, intermediate or operate with virtual assets.[2][3]

That changes how the off-ramp has to be built. The provider needs to know which counterparty can take the stablecoin exposure and which local banking path will hold and move the resulting pesos.

Assuming all these hurdles can be navigated, the final step is landing the fiat in the supplier's account.

USDC → COP

Specific solutions must be found for specific corridors.

Every corridor has different infrastructural and operational needs, reliable execution requires designing systems that support each.

EXHIBIT C4 · FX CONVERSION / OFF-RAMP

From USDC to executable COP

FX ICEBERG

5 Destination Delivery | **4 FX Conversion / Off-Ramp** | 3 Transfer | 2 Compliance Screening | 1 Source Fiat Processing

FX CONVERSION / OFF-RAMP

Provider requests COP quotes	The provider sends requests to the counterparties it can reach in the corridor.
↓ Counterparties return prices	The provider has to determine how much real capacity sits behind those quotes.
↓ Provider assembles enough liquidity	One or several counterparties may be needed to complete the conversion.
↓ COP becomes available	The payout side now has pesos ready for Destination Delivery.

WHERE WEAK INFRASTRUCTURE CAN INTERRUPT THE FLOW

Liquidity Pools

The first counterparty can cover only part of the amount, while the remaining liquidity is unavailable or too expensive.

Technical Infrastructure

The provider cannot reach enough counterparties or distinguish independent capacity from several quotes pointing to the same underlying pool.

Compliance Infrastructure

The local-fiat leg reaches a bank without the information or controls needed to keep it moving, creating another hold or rejection.

Banking Relationships

The resulting COP cannot land in an account the payout side can use to fund the supplier payment.

Regulatory Licenses / Market Access

The off-ramp has to use counterparties and banking roles the Colombian corridor actually permits.[23]

SUPPORTING INFRASTRUCTURE

Banking Relationships | Regulatory Licenses / Market Access | Liquidity Pools | Compliance Infrastructure | Technical Infrastructure

III.06

Destination Delivery

The provider has found the pesos. Now they have to land in the supplier's account.

First, the provider needs a banking or payout partner in Colombia. If that relationship does not already exist, the payment has a much larger problem than a slow payout. New banking relationships can take months to establish.[1] For providers operating in digital assets, banking access has also had a recent history of restrictions and withdrawn services.[2]

Even with a banking partner, the provider may not have direct access to the system that will settle a payment of this size. CUD is Colombia's high-value RTGS system, operated by Banco de la República. Direct participation is limited to authorized institutions with accounts at the central bank, which can use CUD to move funds for themselves or their clients.[3]

Under ordinary conditions, the provider instructs its local partner to make the payment. The bank needs enough pesos in its CUD account to cover the instruction. CUD settles transfers individually and in real time when sufficient funds are available, after which the supplier's bank can credit its customer.[4]

This is where the most moving parts come together. The provider is relying on institutions and systems it does not completely control. Its banking relationship has to remain open. The money has to be in the right place. The payout

instruction has to make it through the local bank and settlement system, and the provider needs to know what happened after it handed the payment off.

Suppose everything upstream has worked. The stablecoins were sourced, the checks cleared, the Transfer completed and the provider found the pesos. It sends the payout instruction and the local partner cannot complete it.

The beneficiary may be held for review.

There may not be enough usable pesos in the right account.

The payment may never reach CUD.

CUD may settle the transfer while the supplier's bank still does not release the money.

Almost the entire transaction can succeed while the payment itself fails.

The provider now has to find out where the pesos actually are and what has to happen next. Strong local relationships matter here. The provider needs to be able to reach the right institution, resolve whatever stopped the payment and understand whether another established payout route can finish the job. Without those relationships, it may simply have nowhere else to go.

None of this can be improvised after the payment breaks. The accounts, counterparties and local access have to exist before the transaction arrives.

The payment is complete when the supplier can use the pesos.

EXHIBIT C5 · DESTINATION DELIVERY

The payment is not finished until the supplier can use the pesos

FX ICEBERG

5 Destination Delivery	4 FX Conversion / Off-Ramp	3 Transfer	2 Compliance Screening	1 Source Fiat Processing

DESTINATION DELIVERY

COP is available	The provider has completed the off-ramp and has pesos ready for payout.
↓ Local partner accepts the instruction	The provider hands the payment to its Colombian banking or payout partner.
↓ CUD settles the payment	The participating institution sends the high-value transfer through Banco de la República.
↓ Supplier receives usable COP	The supplier's bank credits the account. The payment is complete.

WHERE WEAK INFRASTRUCTURE CAN INTERRUPT THE FLOW

Banking Relationships

The provider has no local payout partner, or an existing relationship is restricted or withdrawn. There is no established path into the banking system.

Regulatory Licenses / Market Access

The provider is not a direct CUD participant and has to reach the RTGS system through an authorized institution.

Liquidity Pools

The payout institution does not have enough immediately usable COP in the account needed to settle the instruction.

Compliance Infrastructure

The beneficiary or payment is held for review after the provider has already completed the upstream transaction.

Technical Infrastructure

The provider cannot submit, track or reconcile the payout cleanly enough to know whether the money was rejected, settled or credited.

SUPPORTING INFRASTRUCTURE

Banking Relationships · Regulatory Licenses / Market Access · Liquidity Pools · Compliance Infrastructure · Technical Infrastructure

III.07

Liquidity Is a System

The supplier has the pesos. The payment is complete.

From the Colombian supplier's side, the outcome is almost mundane: money left one place and arrived in their account, the visible tip of the iceberg. The underlying mechanics are only interesting to them when they fail in some way.

This report has been a study of those failures, why they happen and the machinery that must be in place to prevent them.

High-quality liquidity is the capacity to move money from one place to another quickly, reliably and at a reasonable price.

It is a system that must be built corridor by corridor over the course of years, one that must remain adaptive to both the macroeconomic conditions that affect the FX market as a whole and the local conditions that can make a particular transaction more or less difficult to complete.

Stablecoins make it easier to move money in a narrow sense, but transferring coins from wallet to wallet is only one small piece of the stack.

One payment, an entire system underneath it

This is why we developed the FX Iceberg. We think it's the best way to think about the market, especially now as it becomes more dynamic. It helps to illustrate what is really required to move value across corridors: operational capacity and robust infrastructure. The Operations side describes what every transaction must do well to make it to its destination. The Infrastructure supplies the capacities necessary for those Operations to function.

The relationship between the two is many-to-many. A banking relationship can matter when money first enters the transaction and again when it needs to leave. Technical systems support much more than the blockchain. Infrastructural considerations can drive the need to change operational steps, and operational needs can require additional supporting infrastructure.

EXHIBIT C6 · REPORT SYNTHESIS

High-quality liquidity is a system

INFRASTRUCTURE ↓ OPERATIONS →	Source Fiat Processing	Compliance Screening	Transfer	FX Conversion / Off-Ramp	Destination Delivery
Banking Relationships	✓			✓	✓
Regulatory Licenses / Market Access	✓	✓		✓	✓
Liquidity Pools	✓			✓	✓
Compliance Infrastructure		✓		✓	✓
Technical Infrastructure	✓	✓	✓	✓	✓

Liquidity and the market

We began this report with the claim that liquidity sits at the center of FX. It should now be obvious what we meant.

Settlement speed and reliability are dependent on how much high-quality liquidity is available across a corridor. Are the systems capable of providing accurate quotes, filling orders, clearing regulatory hurdles and operating

effectively at larger clip sizes? Are the currency pairs too illiquid for a direct exchange, requiring a vehicle to act as a bridge? How deep are the hedging markets? How robust are the correspondent chains? How many brokers are willing to offer prices at all?

These are different mechanisms responding to different constraints. Together they help explain why the global FX market does not behave like one uniform pool of money.

We broke our analysis into two major “regions,” a High Liquidity World and a Low Liquidity World. This is because the FX market is often inaccurately simplified down to the flows of a handful of high- volume countries. What the market looks like when constrained to USD or EUR is often utterly divorced from how it appears in almost every other corridor on the planet.

Direct FX liquidity is thin	→	Activity routes through deeper common pools, which is why vehicle currencies such as USD become central.
Banking or payment reach is missing	→	Institutions build correspondent relationships, domestic rails or regional systems to extend reach.
Deliverability or hedging is constrained	→	Market activity can migrate into alternate structures such as offshore NDF markets.

For most countries, FX is mired in friction and inefficiencies that slow the flow of funds and have required a significant amount of local innovation to circumvent. If you care about moving money efficiently, as we do, it's these systems that must be understood, particularly when you are trying to move money at scale across a host of market conditions.

Liquidity at any price

That last distinction is essential. Any claim about liquidity depends on the transaction being asked of the market.

CORRIDOR	TRANSACTION SIZE	MARKET CONDITIONS
Where are you moving money?	**How much are you asking the market to carry?**	**What is happening when you need to trade?**
Currency availability, market access, hedge depth and local settlement paths differ from one corridor to another.	Capacity that looks abundant at smaller clips can become scarce or expensive as institutional demand rises.	Liquidity that performs well in ordinary periods can behave differently when volatility, volume or operational stress rises.

This is why institutional liquidity has to be tested with institutional demand:

A market can look deep at $1M and fall apart at $10M.

A corridor can work smoothly in calm conditions and deteriorate the moment interest rates change or a news headline hits.

Even inside a single currency, spot trading can be healthy while hedging or settlement remains constrained.

A system cannot be measured with one headline number

Once liquidity is understood this way, the measurement problem becomes unavoidable. How do you know if the market is healthy? The headline numbers, trading volume, reserve size, quoted spreads and rail availability, can help at a glance, but it should be clear now that they aren't the entire picture.

This is why our Liquidity Scorecard asks a different question: how does the liquidity behave when someone actually tries to use it?

EXHIBIT C7 · SCORECARD SYNTHESIS

Measure the outcome of the system

Can you transact as expected?	Spread Consistency: does the offered price remain usable? Execution Quality: do you get the trade you expected? Rejection / Last Look: does apparently available liquidity disappear when you try to use it?
Does the liquidity remain usable?	Depth-of-Book Consistency: is meaningful capacity still there behind the first trade? Price Accuracy / Alignment: does the executable price remain sensible against a relevant alternative?
Does the money actually arrive?	Settlement Speed Under Load: does the complete payment finish reliably as demands on the system rise?

The Scorecard measures the outcome. The Iceberg shows the capacity required to create it. Together they give us a way to evaluate high-quality liquidity without pretending one number can describe the whole transaction.

A strong reading in one place cannot carry the rest of the system. Tight pricing can sit on top of shallow depth. A successful execution can still end in slow delivery. The quality of the liquidity emerges from how these outcomes hold together over time.

Someone has to build the capacity

For a buyer, our Scorecard is meant to act as a diligence tool. For a builder, these same measurements describe the outcomes your product has to reproduce.

In both cases, the analysis and the work are corridor-specific, and so are the questions you need to answer:

Where is usable capital?	What amount is actually available in the currencies and forms needed at the sizes customers will transact?
Who can move it?	Which counterparties and financial institutions can perform the required roles along the route?
What does the corridor permit?	Which regulatory and market-access constraints determine how the transaction can be structured?
How does the money reach the recipient?	Which local banking, settlement and payout systems complete the last part of the transaction?
What happens when the expected route fails?	Which alternatives already exist, and can the provider identify where the money is while the problem is being resolved?

This is why local knowledge matters. Different corridors find their friction in different places. A provider that understands those constraints can build around them before volume arrives. A provider that assumes the same architecture will work everywhere will discover problems only when the system is most under strain.

The machinery will keep changing

Stablecoins have already changed an important part of the architecture. Value can move across a blockchain continuously, using a shared ledger and largely eliminating the need for correspondent- level reconciliation during the transfer itself. This has dramatically changed the reconciliation process and the orchestration layer that sits around the transfer.

Software will continue to absorb more of the coordination. AI and automation can help providers search for routes, manage inventory, monitor transactions and respond to exceptions while money is still in flight.

Those tools are not replacements for good operations or good infrastructure. They magnify the capabilities available to providers. They do not manufacture local capacity where none exists. A system can choose among available routes more intelligently; it still needs a viable route to choose.

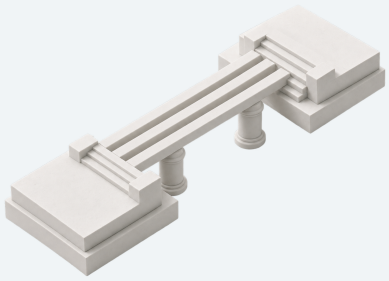
The underlying test therefore remains the same: can you move the amount required, at the expected price, and deliver usable money reliably?

The High Liquidity World is not a closed club confined to the G7. More markets can move toward it as the systems that support high-quality liquidity become available to them.

The opportunity exists to extend these capacities into more corridors. Stablecoins provide a faster transfer layer, while better software makes more of the surrounding complexity manageable. The institutions that understand the Iceberg and the Scorecard, and know how to assemble all the moving parts into a stable system for moving money across the globe, will succeed.

Our goal is to make money move across the world as quickly as data. That requires technology capable of coordinating the transaction and enough real liquidity underneath it to make the transaction possible.

We are building that at OpenFX, and we want to build a world where those rails are available to everyone.



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